

GE Healthcare

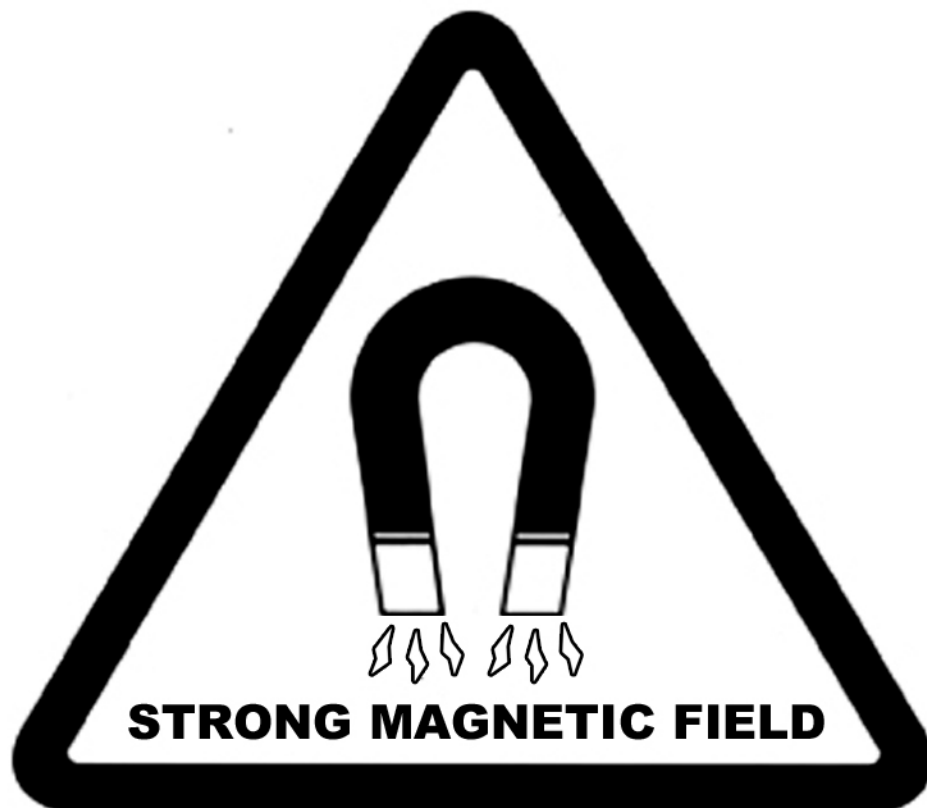
Oxygen Monitor Subsystem Manual



OPERATING DOCUMENTATION

46-015336
Revision 4

WARNING



NO PACEMAKERS NO METALLIC IMPLANTS

Persons with pacemakers, neurostimulators or metallic implants must not enter the magnet area. Serious injury may result.



NO LOOSE METAL OBJECTS

Iron and steel materials must not be taken into the magnet area. Serious injury or property damage may result.

Important Information

LANGUAGE

ПРЕДУПРЕЖДЕНИЕ (BG)	Това упътване за работа е налично само на английски език. • Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод. • Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа. • Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.
警告 (ZH-CN)	本维修手册仅提供英文版本。 • 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。 • 未详细阅读和完全理解本维修手册之前，不得进行维修。 • 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。
警告 (ZH-HK)	本服務手冊僅提供英文版本。 • 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。 • 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。 • 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。
警告 (ZH-TW)	本維修手冊僅有英文版。 • 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。 • 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。 • 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。
UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. • Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. • Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. • Zanimarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.

**VÝSTRAHA
(CS)**

Tento provozní návod existuje pouze v anglickém jazyce.

- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

**ADVARSEL
(DA)**

Denne servicemanual findes kun på engelsk.

- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

**WAARSCHUWING
(NL)**

Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.

- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

**WARNING
(EN)**

This service manual is available in English only.

- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

**HOIATUS
(ET)**

See teenindusjuhend on saadaval ainult inglise keeles.

- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

**VAROITUS
(FI)**

Tämä huolto-ohje on saatavilla vain englanniksi.

- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

**ATTENTION
(FR)**

Ce manuel d'installation et de maintenance est disponible uniquement en anglais.

- Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

**WARNUNG
(DE)**

Diese Serviceanleitung existiert nur in englischer Sprache.

- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendienst-technikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

**ΠΡΟΕΙΔΟΠΟΙΗΣΗ
(EL)**

Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά.

- Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

**FIGYELMEZTETÉS
(HU)**

Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.

- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

**AÐVÖRUN
(IS)**

Þessi þjónustuhandbók er aðeins fánleg á ensku.

- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

**AVVERTENZA
(IT)**

Il presente manuale di manutenzione è disponibile soltanto in lingua inglese.

- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

**警告
(JA)**

このサービスマニュアルには英語版しかありません。

- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

**경고
(KO)**

본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다.

- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오.
- 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

**BRĪDINĀJUMS
(LV)**

Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā.

- Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu.
- Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumų risks apkopes sniedzējam, operatoram vai pacientam.

**ĮSPĖJIMAS
(LT)**

Šis eksploatavimo vadovas yra tik anglų kalba.

- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas.
- Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo.
- Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

**ADVARSEL
(NO)**

Denne servicehåndboken finnes bare på engelsk.

- Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

**OSTRZEŻENIE
(PL)**

Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.

- Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

**ATENÇÃO
(PT-BR)**

Este manual de assistência técnica encontra-se disponível unicamente em inglês.

- Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

**ATENÇÃO
(PT-PT)**

Este manual de assistência técnica só se encontra disponível em inglês.

- Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

**ATENȚIE
(RO)**

Acest manual de service este disponibil doar în limba engleză.

- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

**ОСТОРОЖНО!
(RU)**

Данное руководство по техническому обслуживанию представлено только на английском языке.

- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

**UPOZORENJE
(SR)**

Ovo servisno uputstvo je dostupno samo na engleskom jeziku.

- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodi-lačke usluge.
- Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.

**UPOZORNENIE
(SK)**

Tento návod na obsluhu je k dispozícii len v angličtine.

- Ak zákazník poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.

**ATENCION
(ES)**

Este manual de servicio sólo existe en inglés.

- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

**VARNING
(SV)**

Den här servicehandboken finns bara tillgänglig på engelska.

- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

**OPOZORILO
(SL)**

Ta servisni priročnik je na voljo samo v angleškem jeziku.

- Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod.
- Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli.
- Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.

**DİKKAT
(TR)**

Bu servis kılavuzunun sadece ingilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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Revision History

Revision	Date	Reason for Change
4	July 13, 2010	Updated per PCN 304036. Formatted in eCLS, the content management tool, as a publication, and added new model, ISA-60M.
3	July 10, 1997	Rewritten to reflect aerosol calibration procedure and 5 year oxygen sensor.
2	March 14, 1994	Correct part number errors and remove Omission/Error Report (Direction 15028).
1	June 21, 1993	Rewritten to reflect changes in hardware.
0	May 31, 1990	Initial Release.

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Chapter 1 Damage in Transportation

1 Damage in Transportation

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 6.
- Fill out a report on <http://egems.med.ge.com/edq/home.jsp>
- Contact your local service coordinator for more information on this process.

Rev. June 13, 2006

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Chapter 2 Introduction

1 Introduction

1.1 Overview

NOTE: This manual's vendor number is 15336.



WARNING

CRYOGEN GAS MAY BE GIVEN OFF IN THE MAGNET ROOM DURING FILLING AND RAMPING OF THE MAGNET, DISPLACING THE OXYGEN IN THE ROOM. TO PREVENT ASPHYXIATION, THE OXYGEN MONITOR SUBSYSTEM MUST BE INSTALLED, CALIBRATED, AND OPERATIVE PRIOR TO MOVING THE MAGNET INTO THE MAGNET ROOM.

This manual contains procedures on how to install, calibrate, and maintain the Oxygen Monitor Subsystem (Model ISA-42M and Model ISA-60M).

Illustration 2-1: Model ISA-42M



Illustration 2-2: Model ISA-60M

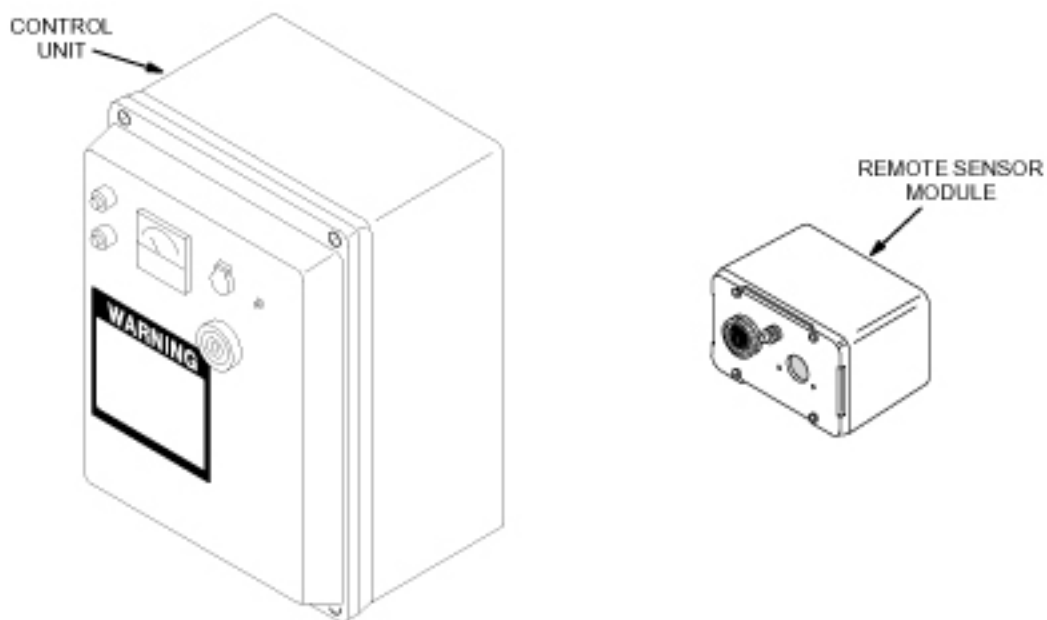


The Oxygen Monitor subsystem monitors the oxygen level in environments where oxygen depletion can occur, such as an MR Magnet Room. The subsystem activates audible and visual alarms when oxygen levels drop below safe levels.

The major components in the Oxygen Monitor subsystem are shown and described below:

- Control Unit: Contains most of the signal processing circuitry, the power supply, the meter, and audio and visual alarms. The Control Unit is typically mounted in the Operator Room near the Operator Console.
- Remote Sensor Module: Includes an audio and visual alarm and an Oxygen Sensor Cell. The Remote Sensor Module is typically mounted in the Magnet Room.

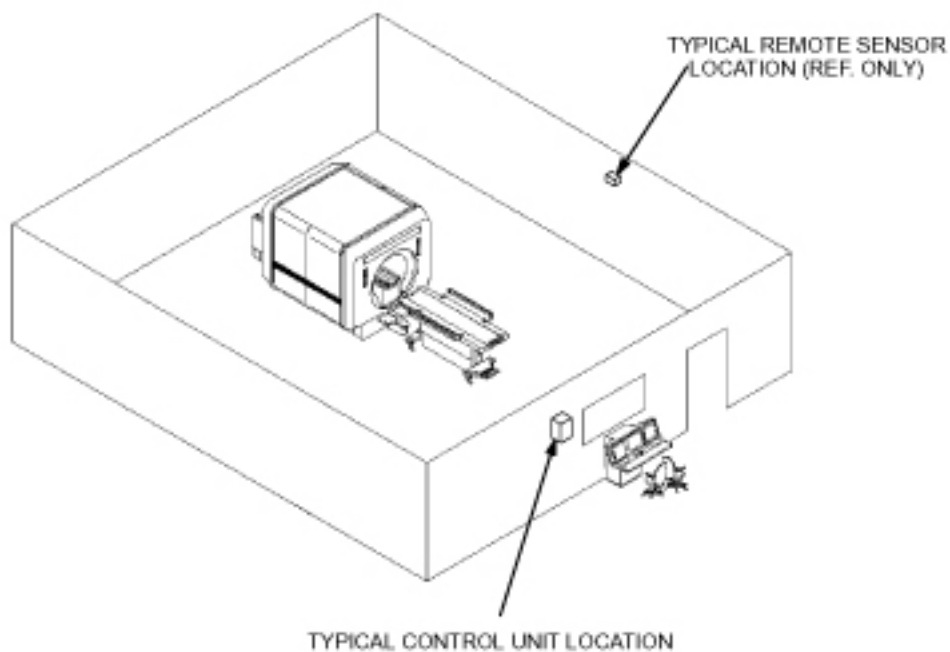
Illustration 2-3: Major Components in Oxygen Monitor Subsystem



NOTE: Cellular phones can cause false alarms in the Oxygen Monitor subsystem. Keep all cellular phones at least 20 feet away from Oxygen Monitor subsystems even if the phone is not in use.

The illustration below shows the Oxygen Monitor subsystem installation at a typical site.

Illustration 2-4: Typical Oxygen Monitor Subsystem Installation



1.2 What This Manual Contains

- [Chapter 3, Theory](#), describes how the Oxygen Monitor subsystem works.
- [Chapter 4, Installation](#), contains step-by-step procedures for installing the Oxygen Monitor subsystem.
- [Chapter 5, Calibration](#), contains step-by-step procedures for calibrating the Oxygen Monitor subsystem.
- [Chapter 6, Replacement](#), contains step-by-step procedures for replacing Field Replaceable Units (FRUs) of the Oxygen Monitor subsystem.
- [Chapter 7, Renewal Parts](#), contains parts lists identifying the parts and part numbers that are replaceable in the Oxygen Monitor subsystem.

Chapter 3 Theory

1 Theory

NOTE: This chapter was written for the ISA-42M Oxygen Monitor only.

The Oxygen Monitor subsystem senses the presence of oxygen with a galvanic micro-fuel cell. The cell produces an output current proportional to the partial pressure of oxygen in the surrounding atmosphere. The partial pressure corresponds to the percentage of oxygen by volume at normal atmospheric pressure. Circuitry within the Oxygen Monitor Control Unit converts the cell's output current to a value representing the percentage of oxygen by volume and displays the percentage on a meter. The meter has a range of 16 to 26%.

Typically, air contains 20.9% oxygen by volume. Inert gas, such as helium or nitrogen, released within an enclosed environment can displace the air within the environment and reduce oxygen levels to the point where the environment becomes hazardous to human life. When the oxygen level drops below 18%, the Oxygen Monitor subsystem does the following:

- Activates an audible alarm.
- Illuminates a warning light.
- Displays the percentage of oxygen by volume on a meter.
- Activates an optional exhaust fan.

The Oxygen Monitor subsystem system consists of a:

- Control Unit.
- Remote Sensor Module.

1.1 Control Unit

The Control Unit contains:

- Circuitry that processes signals from the Remote Sensor Module.
- The power supply which supplies power for the Control Unit and the Remote Sensor Module.
- Circuit breakers and fuses for electrically protecting the Oxygen Monitor subsystem.
- Alarm LEDs and horn to indicate an alarm condition.
- A meter that displays the oxygen level in percent by volume.
- Relays to activate optional equipment, such as an exhaust fan, when the Oxygen Monitor subsystem enters an alarm condition.

The Control Unit receives a voltage from the Oxygen Sensor Cell in the Remote Sensor Module proportional to the percentage of oxygen by volume in the magnet room. Signal processing circuitry

within the Control Unit amplifies this voltage and applies it to the meter. The gain potentiometer adjusts the amplification level to produce an accurate meter reading. Other circuits activate the alarm horn and LEDs or energize optional relays when the amplified voltage exceeds a preset threshold value. Potentiometers on the Main Circuit board set these thresholds. Refer to [Chapter 5, Calibration](#) for procedures to set and calibrate the thresholds.

1.1.1 Power

External power is supplied to the entire Oxygen Monitor subsystem through the power supply within the Control Unit. Two circuit breakers and a slow-blow fuse protect the Oxygen Monitor subsystem.

1.1.2 Relays (Optional Equipment)

1.1.2.1 For Model ISA-42M

Two relays are available in the Control Unit which can be used to automatically activate customer-provided equipment, such as an exhaust fan, when the Oxygen Monitor subsystem enters an alarm state. The relay contacts are dry, with a rating of 2 amps continuous, 5 amps surge, at 120 VAC. One of the relays is a latching relay, the other is a non-latching relay.

The latching relay is available at TB-7. Once triggered by an alarm condition, this relay remains latched (even when the alarm condition is no longer present) until it is released by pushing the RELAY RELEASE pushbutton switch on the Control Unit front panel. This relay is not fail-safe; the power-off position of the contacts is opposite from the latched (alarm) position. If power to the Control Unit is interrupted, this relay will not be energized. The contact positions for the power-off, non-alarm condition are shown in the table below.

Table 3-1: Latching Relay Contacts (TB-7)

Terminal Label	Contact Description
TB7-1NO	Normally Open
TB7-1NC	Normally Closed
TB7-1C	Common

The non-latching relay is available at TB3. This is a fail-safe relay; the power-off position of the contacts is the same as the alarm position. This relay is only energized during an alarm condition or when power is interrupted. This is the relay that is normally used to activate a customer-provided exhaust fan (refer to Optional Equipment Installation section). The contact positions for the power-off, alarm condition are shown in the table below.

Table 3-2: Non-Latching Relay Contacts (TB-3)

Terminal Label	Contact Description
TB3-1NO	Normally Open
TB3-1NC	Normally Closed
TB3-1C	Common

1.1.2.2 For Model ISA-60M

See the vendor manual.

1.2 Remote Sensor Module

The Remote Sensor Module measures the oxygen level in the magnet room, relays the measurement to the Control Unit, and warns occupants of the magnet room of a low oxygen condition. The Remote Sensor Module contains:

- An alarm LED and horn.
- Oxygen Sensor Cell (18-month/5-year)

The Remote Sensor Module receives power from the Control Unit to operate the alarm horn and LED. The Oxygen Sensor Cell produces the voltage indicating the oxygen level.

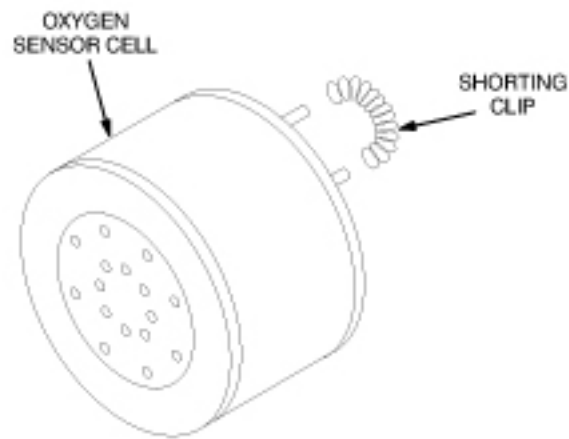
1.2.1 Oxygen Sensor Cell (Model ISA-42A)

The Oxygen Sensor Cell is a galvanic micro-fuel cell. Oxygen in the gas space surrounding the cell diffuses through a Teflon membrane and is reduced on the surface of the cathode. A corresponding oxidation occurs at the anode which generates an electrical current proportional to the concentration of oxygen. Cell output is limited by the rate at which oxygen enters the cell and by the amount of anode material stored within the cell. The resulting cell output drives the meter circuit of the Oxygen Monitor Control Unit.

1.2.2 Oxygen Sensor Cell Shorting Clip (Model ISA-42A)

The shorting clip on the Oxygen Sensor Cell is removed immediately prior to installation as shown below. During storage, the shorting clip provides a current loop assuring that the sensor cell is ready for immediate use when placed into an Oxygen Monitor subsystem. If the shorting clip is prematurely removed, oxygen diffuses into the cell until the electrolyte is saturated. In this condition, it may be necessary to wait a considerable length of time before placing the cell into an operating situation. If the shorting clip is removed from a cell for some time, it will drive the meter off scale when it is installed in the Oxygen Monitor subsystem. This condition corrects itself in time, or the installer can remove the cell and place a shorting clip on the cell. For every hour the cell is un-shortened, about one hour is needed to re-stabilize the cell.

Illustration 3-1: Oxygen Sensor Cell



1.2.3 18-Month Oxygen Sensor Cell Characteristics (Model ISA-42A)

The Oxygen Sensor Cell (C-3) has the following characteristics:

- Useful life of the cell depends upon the length of time it is exposed to oxygen and the magnitude of the oxygen concentration. In air, the cell will provide about 18 months of operation. If left in its Environmental Gas Barrier Bag, the cell shelf life is about 3 years.
- Output at 25° C: 200 micro-amperes.
- Operating temperature range: 32-131° F.
- Minimum temperature: 26° F.
- Position Sensitivity: Typically 2 to 5% of reading while mobile van is in transit, returning to calibrated specification after vehicular motion ceases.

1.2.4 5-Year Oxygen Sensor Cell Characteristics (Model ISA-42A)

The 5-year Oxygen Sensor consists of a sensor cell, a battery, and a PC board (with conversion circuitry contained in an RF-shielded enclosure). Incorporated with this new cell is an electronic circuit which duplicates the signal response of the original cell, thereby affording you a direct replacement component. (Aside: The incorporation of a battery was required since power is not available at the Remote Sensor Module in the magnet room.) Operating at less than 30 microamps, this Oxygen Sensor is optimized for a 5 year design life.

- Output voltage in air, 7mV to 15mV at 21% O₂.
- Operating temperature range: 41-104° F.
- Minimum temperature: 41° F.

Chapter 4 Installation

1 Installation

1.1 Model ISA-42M

NOTE: See [Section 1.2](#) for installation process for the ISA-60M Oxygen Monitor.

1.1.1 Oxygen Monitor Kit Contents



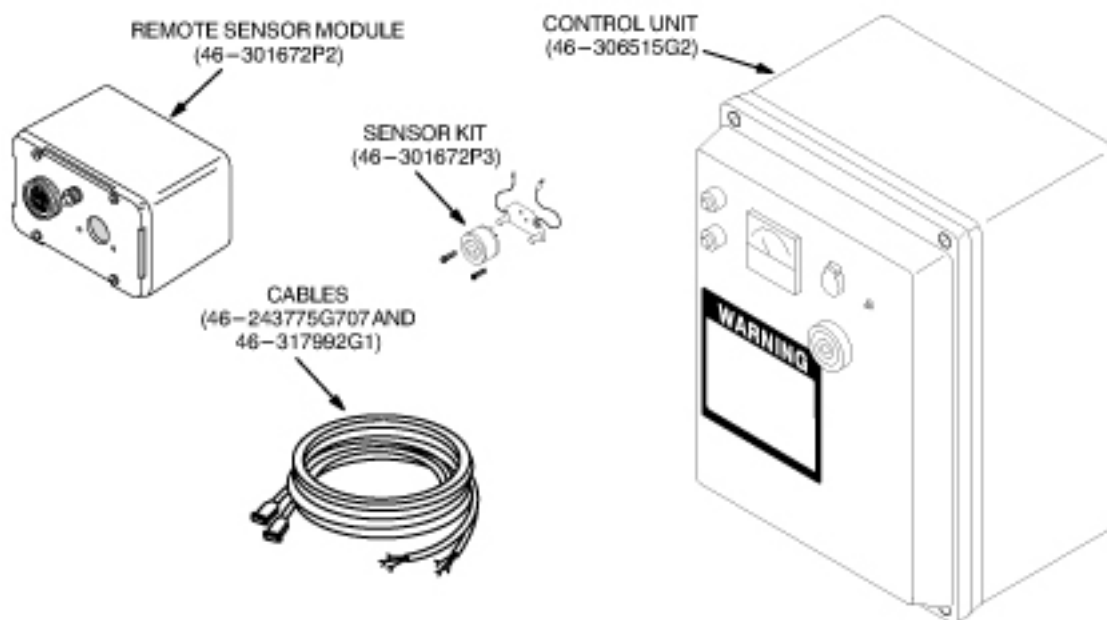
NOTICE

The Oxygen Monitor Subsystem must be installed and functional prior to moving the magnet into the magnet room.

The Oxygen Monitor Kit (46-317271G1) contains the parts needed to assemble an Oxygen Monitor subsystem. See below. The Oxygen Monitor Kit contains the following:

- Control Unit (46-306515G2) (miscellaneous hardware (Ty-wraps and mounting screws) shipped inside Control Unit).
- Remote Sensor Module (46-301672P2).
- Sensor Kit (46-301672P3).
- Cables (46-243775G707 and 46-317992G1).
- Direction 15336, Oxygen Monitor Subsystem.

Illustration 4-1: Oxygen Monitor Subsystem Kit



1.1.2 Installation Procedures

NOTE: The procedures in this section are only applicable to the current Oxygen Monitor subsystem delivered with forward production systems.

1.1.2.1 Installing the Remote Sensor Module



NOTICE

If drilling into RF shielding, holes for mounting screws must be small enough to insure that screws provide good electrical contact with conductive materials in RF shielding.



WARNING

BURN HAZARD! IF THE OXYGEN SENSOR CELL IS LEAKING OR PUNCTURED FOR ANY REASON, PLACE THE CELL ASIDE AND WASH HANDS AND ANY OTHER PARTS OF BODY OR CLOTHING THAT MIGHT HAVE COME IN CONTACT WITH THE ELECTROLYTE LIQUID (KOH SOLUTION). USE LARGE QUANTITIES OF WATER; THE ELECTROLYTE IS A STRONG CAUSTIC SOLUTION AND CAN CAUSE SEVERE BURNS.



NOTICE

If electrolyte liquid or crystal growth are visible anywhere on the cell or within the barrier bag, the cell is leaking and should not be used.

1. Loosen the four screws securing the front cover of the Remote Sensor Module and lift off the front cover. Refer to [Illustration 4-3](#).
2. Mount the Remote Sensor Module in the magnet room according to the site plan. [Illustration 4-4](#) shows the typical installation.

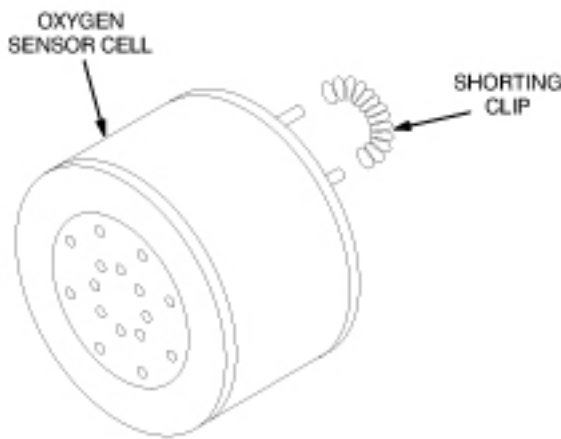
NOTE: The Oxygen Monitor remote sensor will be provided by General Electric. The center of the remote sensor is to be 60 inches from the magnet room floor. The horizontal mounting position must be outside the 100 Gauss field.

- The preferred location of the remote sensor is near the patient entrance on the inside of the magnet room.
- Refer to the architect's layouts to determine the specific location.

All mounting hardware must be non-magnetic.

3. Remove the Shorting Clip on the Oxygen Sensor Cell just prior to installation. See below.

Illustration 4-2: Oxygen Sensor Cell



4. Remove the Oxygen Sensor Cell from the environmental gas barrier bag and remove the shorting clip. See [Illustration 4-5](#).
 5. Install the cell into the Sensor Circuit Board making sure the cell's center pin is inserted into the center socket of the Sensor Circuit Board. See [Illustration 4-6](#).
- NOTE:** New Remote Sensor Modules may include a Sensor Retrofit Plate, which is used with the new Oxygen Calibration Kit. If the Sensor Retrofit Plate is not included, ignore the references to the plate in the following note and step. To obtain the Sensor Retrofit Kit, order Part Number 46-301672P3. For installation instructions, refer to [Section 1.1.4](#).
- NOTE:** During installation of the Sensor Retrofit Plate, make sure the gasket side of the plate is installed against the sensor cell. Also, due to variations in sensor circuit board design, two sets of screws are provided in the kit. Be sure to use the appropriate length screws so that the sensor cell and circuit board are securely fastened to the sensor retrofit plate.
6. Install the Oxygen Sensor Cell, Sensor Circuit Board, and Sensor Retrofit Plate on the front cover with the two screws provided.
 7. Connect the Sensor Circuit Board green wire to terminal 4 in the Remote Sensor Module and connect the orange wire to terminal 5. See [Illustration 4-7](#).
 8. Connect the five wires from cable Run #458 to terminal 1 through 5 as shown in [Illustration 4-7](#).
 9. Secure the cable to the cable mount with one of the Ty-wraps provided in the Oxygen Monitor kit.
 10. Install and secure the front cover of the Remote Sensor Module.

Illustration 4-3: Removing The Remote Sensor Module Cover

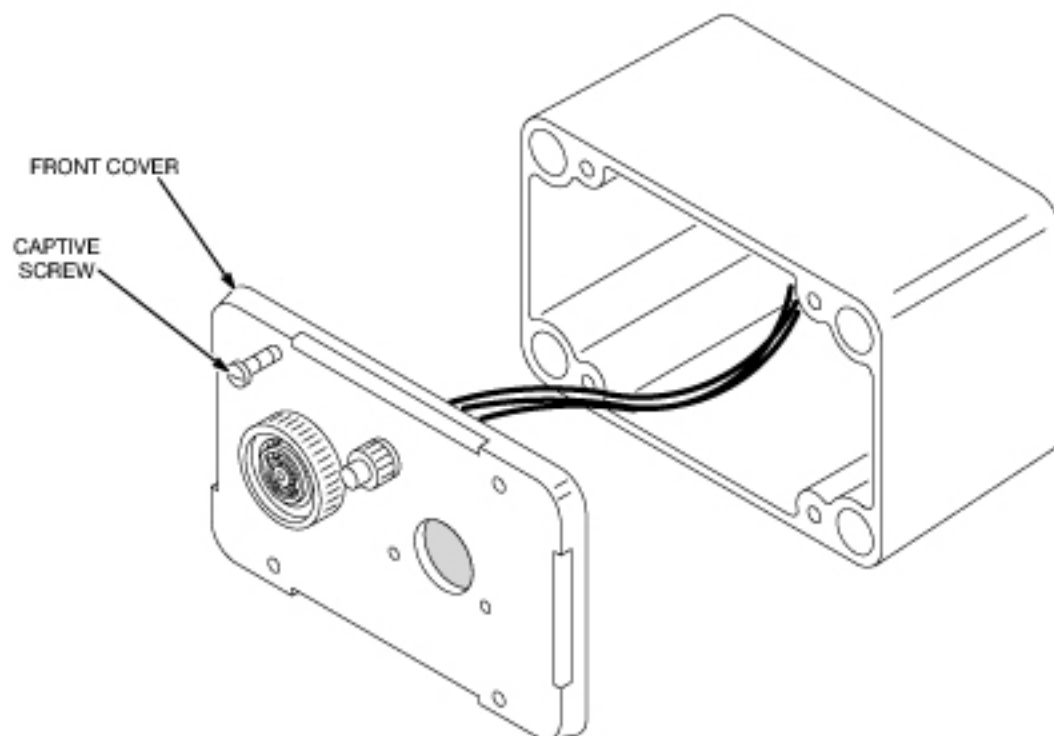


Illustration 4-4: Remote Sensor Module Typical Installation

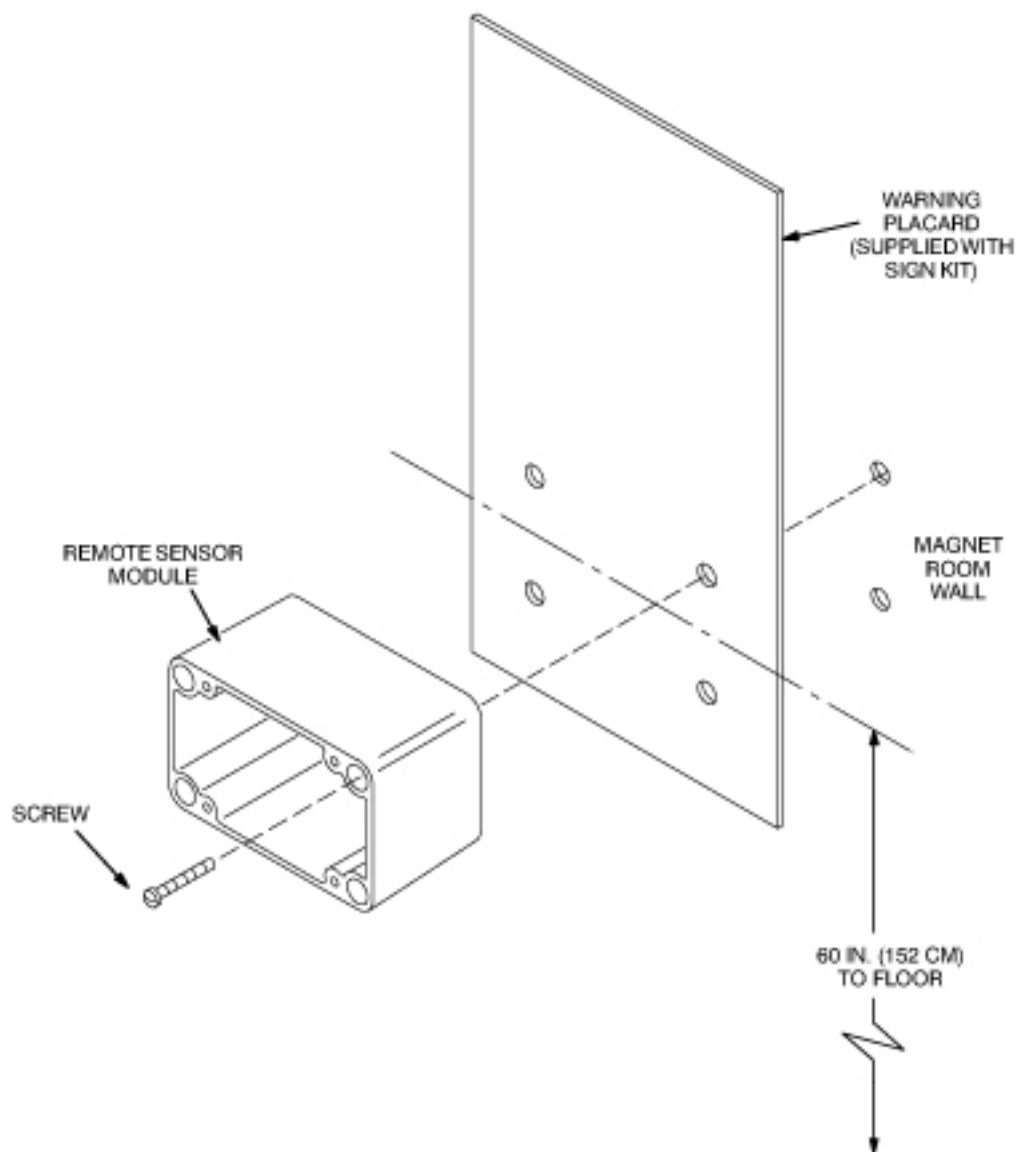


Illustration 4-5: Unpacking The Oxygen Sensor Cell

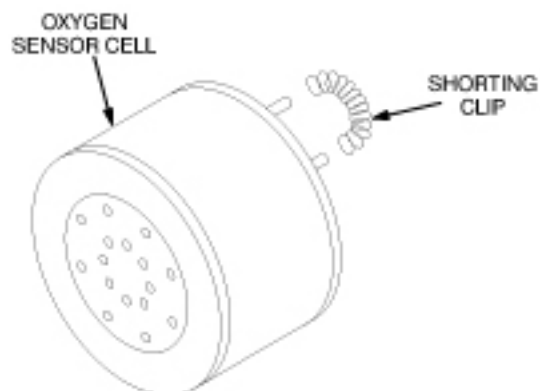


Illustration 4-6: Installing The Oxygen Sensor Cell

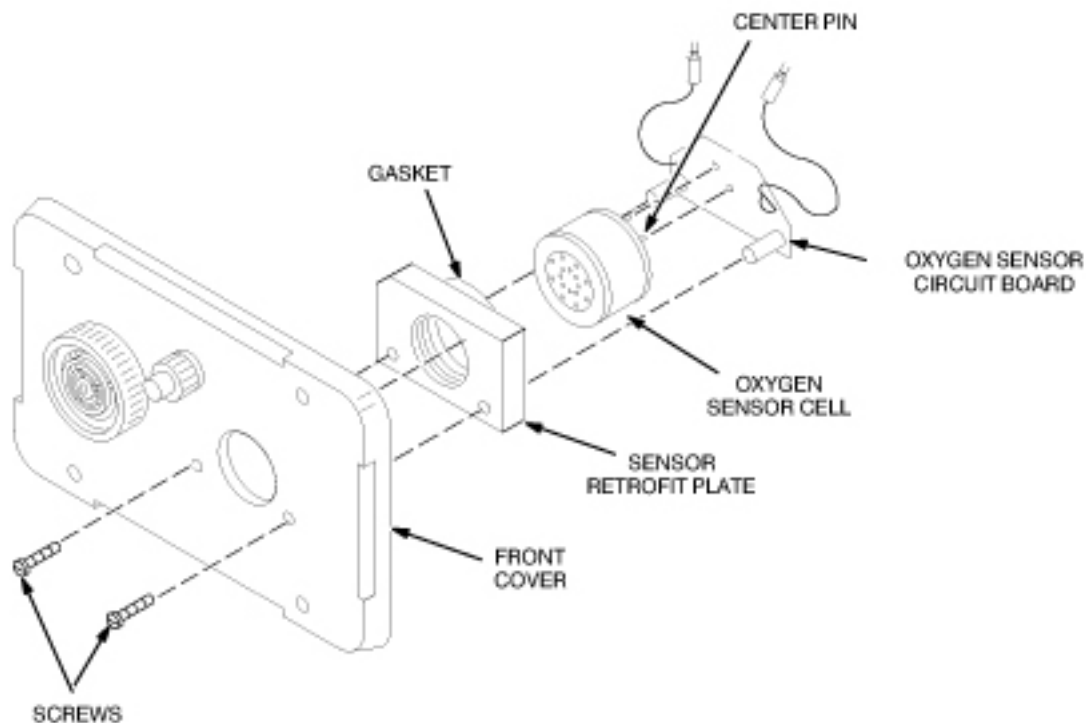
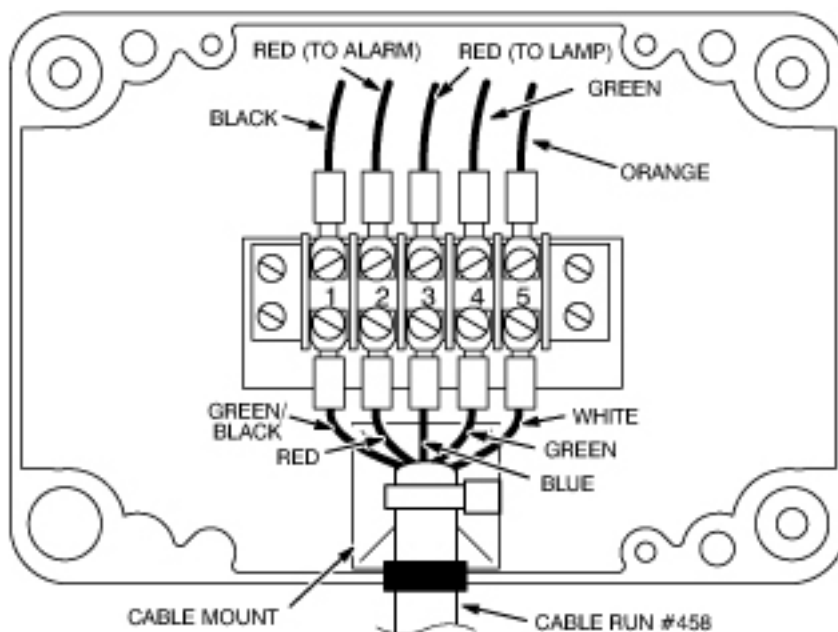


Illustration 4-7: Remote Sensor Module Wiring



1.1.2.2 Installing the Control Unit

**WARNING**

ELECTRICAL SHOCK HAZARD!

TO PREVENT POSSIBLE ELECTRIC SHOCK, LOCATE THE AUXILIARY POWER DISCONNECT SWITCH FOR THE OXYGEN MONITOR SUBSYSTEM, AND SET THE SWITCH IN THE OFF POSITION.

TAG THE SWITCH, INDICATING SERVICE IS BEING PERFORMED AND NOT TO ENERGIZE UNTIL THE SIGNER ON THE TAG IS NOTIFIED OR THE TAG IS REMOVED.

**NOTICE**

Input power circuitry and wiring must meet all applicable provisions of the national electrical code. Also, do not inadvertently interchange the neutral (white) and hot (black) wires at the terminal block. Serious degradation of the instrument performance can result from this error.

1. Loosen the four captive screws securing the front cover of the Control Unit and open the front cover. See [Illustration 4-8](#).
2. Mount the Control Unit in the Operator Room according to the site plan. [Illustration 4-8](#) shows a typical installation.

NOTE: The Oxygen Monitor Control Unit will be provided by General Electric.
The Oxygen Monitor Control Unit should be located near the Operator Console.
Refer to the architect's layouts to determine the specific location.

3. Set up the Control Unit transformer configuration for either 110 VAC or 220 VAC as shown in [Illustration 4-9](#).
4. Turn off external power.
5. Connect wiring as shown in [Illustration 4-10](#) and [Illustration 4-11](#).
6. Secure cable Run #457 to the cable mount with one of the Ty-wraps provided in the Oxygen Monitor Subsystem kit.
7. Close and secure the front cover of the Control Unit.

Illustration 4-8: Control Unit Installation

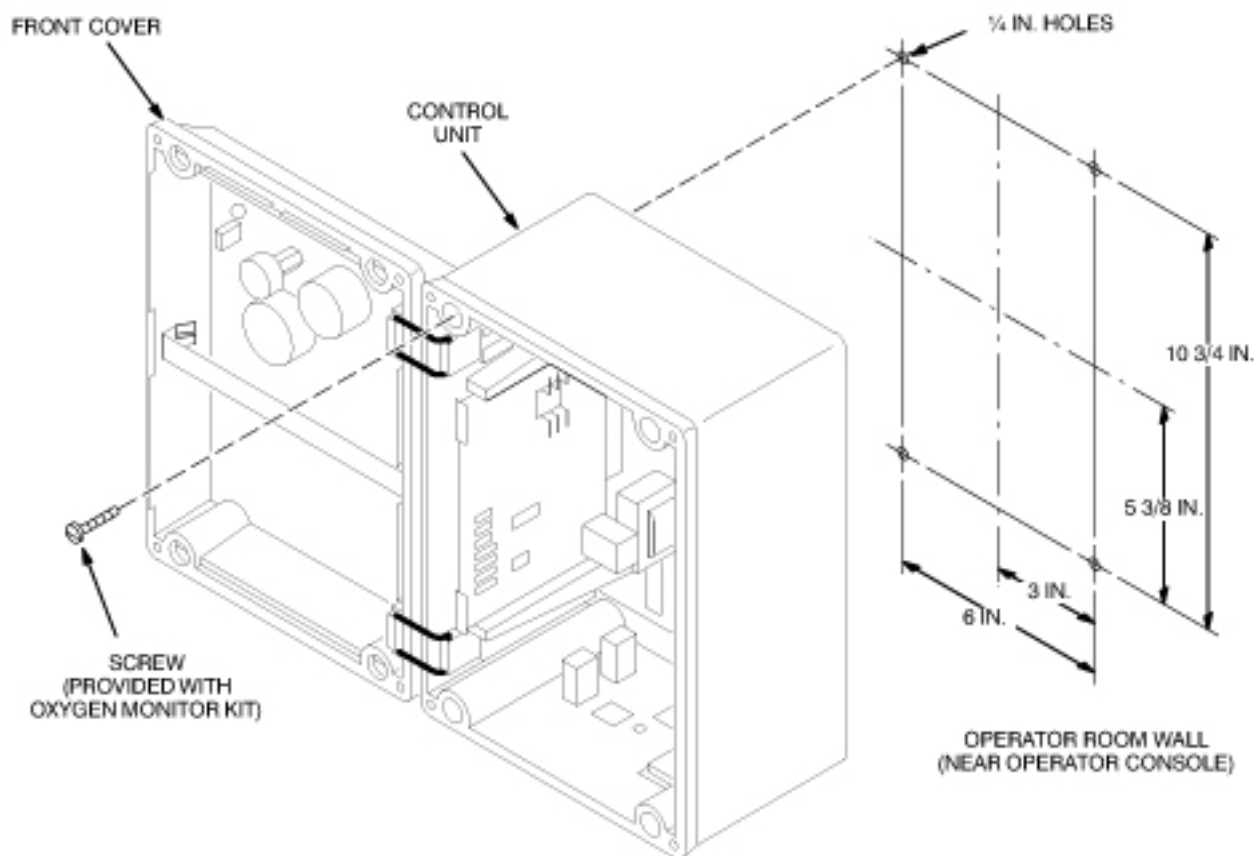


Illustration 4-9: Transformer Configuration

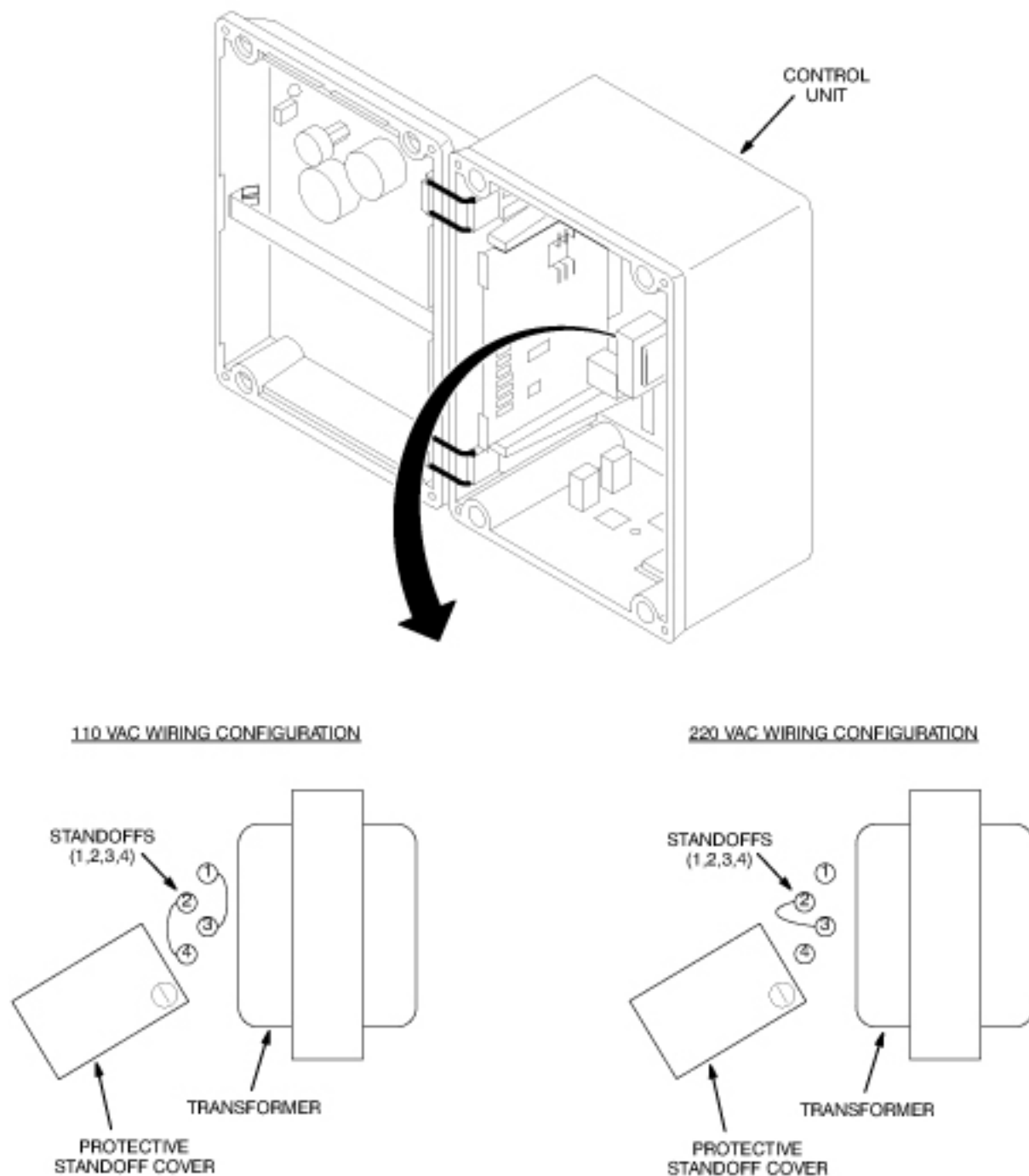
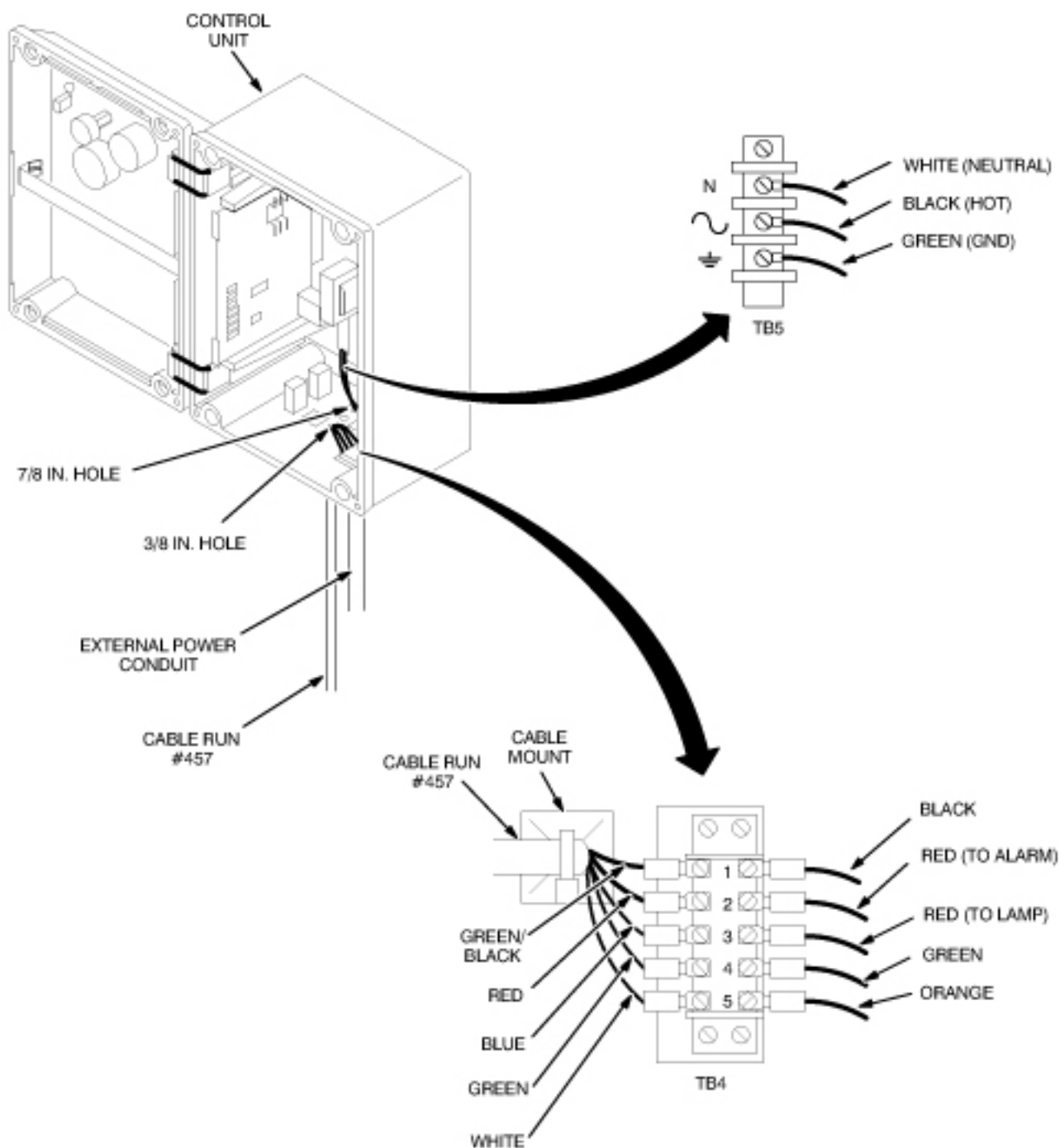


Illustration 4-10: Control Unit Terminal Block Wiring

CABLE	WIRE	TERMINAL	TERMINAL BLOCK
Run #457	Green/Black	1	TB4
	Red	2	
	Blue	3	
	Green	4	
	White	5	
External Power	White (Neutral)	N	TB5
	Black (Hot)	~	
	Green (Ground)	⏏	

Illustration 4-11: Control Unit Wiring



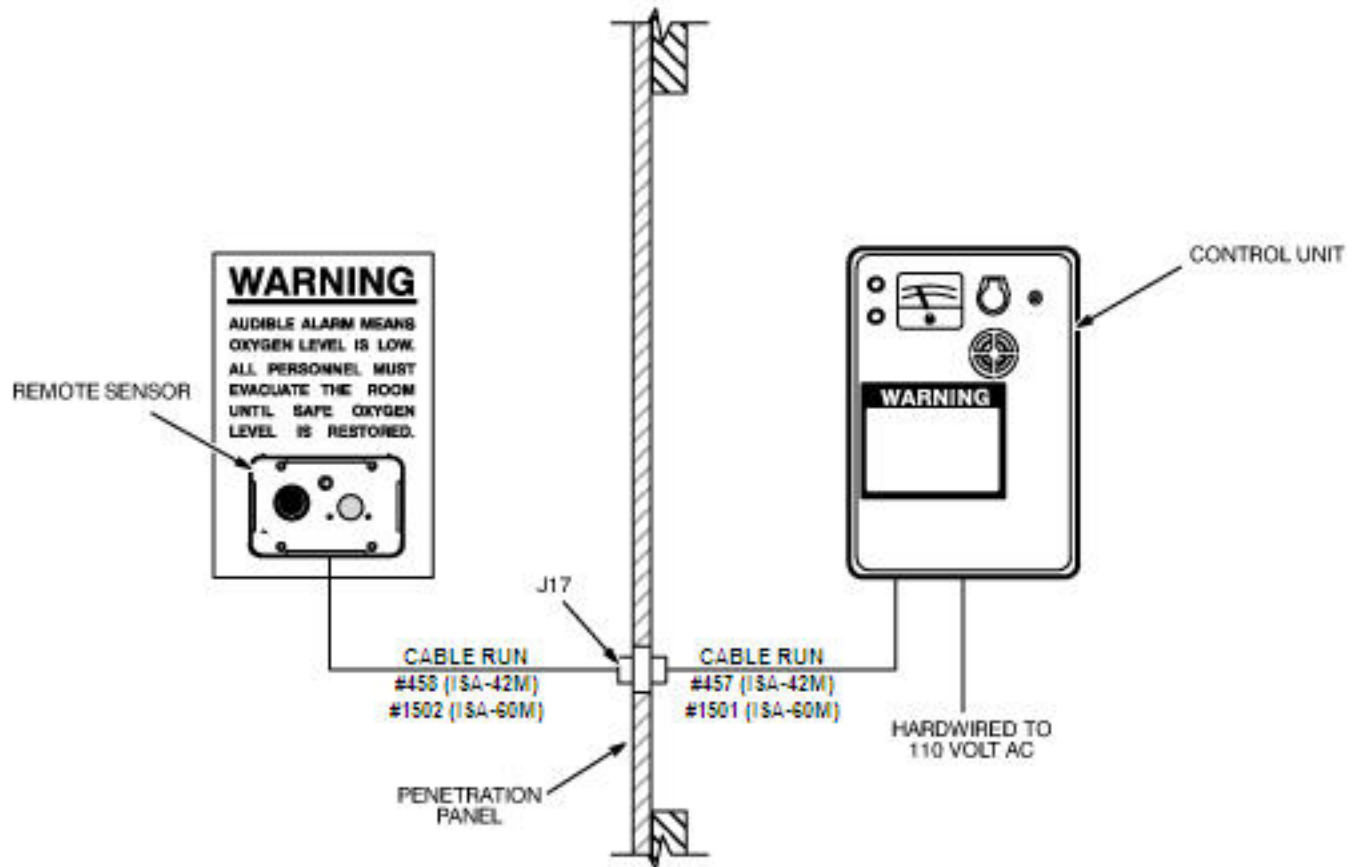
1.1.2.3 Connecting the Remote Sensor Module to the Control Unit

1. Route cable run #458 (ISA-42M) or #1502 (ISA-60M) from the Remote Sensor Module to the Penetration Panel or Secondary Penetration Wall (SPW) and connect the cable to J17. See below.

NOTE: Make sure the extra cable length from cable Run #458 is located away from the magnet to prevent it from causing MR image quality problems.

2. Route cable run #457 (ISA-42M) or #1501 (ISA-60M) from the Control Unit to the Penetration Panel or SPW and connect the cable to J17. See below.

Illustration 4-12: Connecting the Remote Sensor Module to the Control Unit

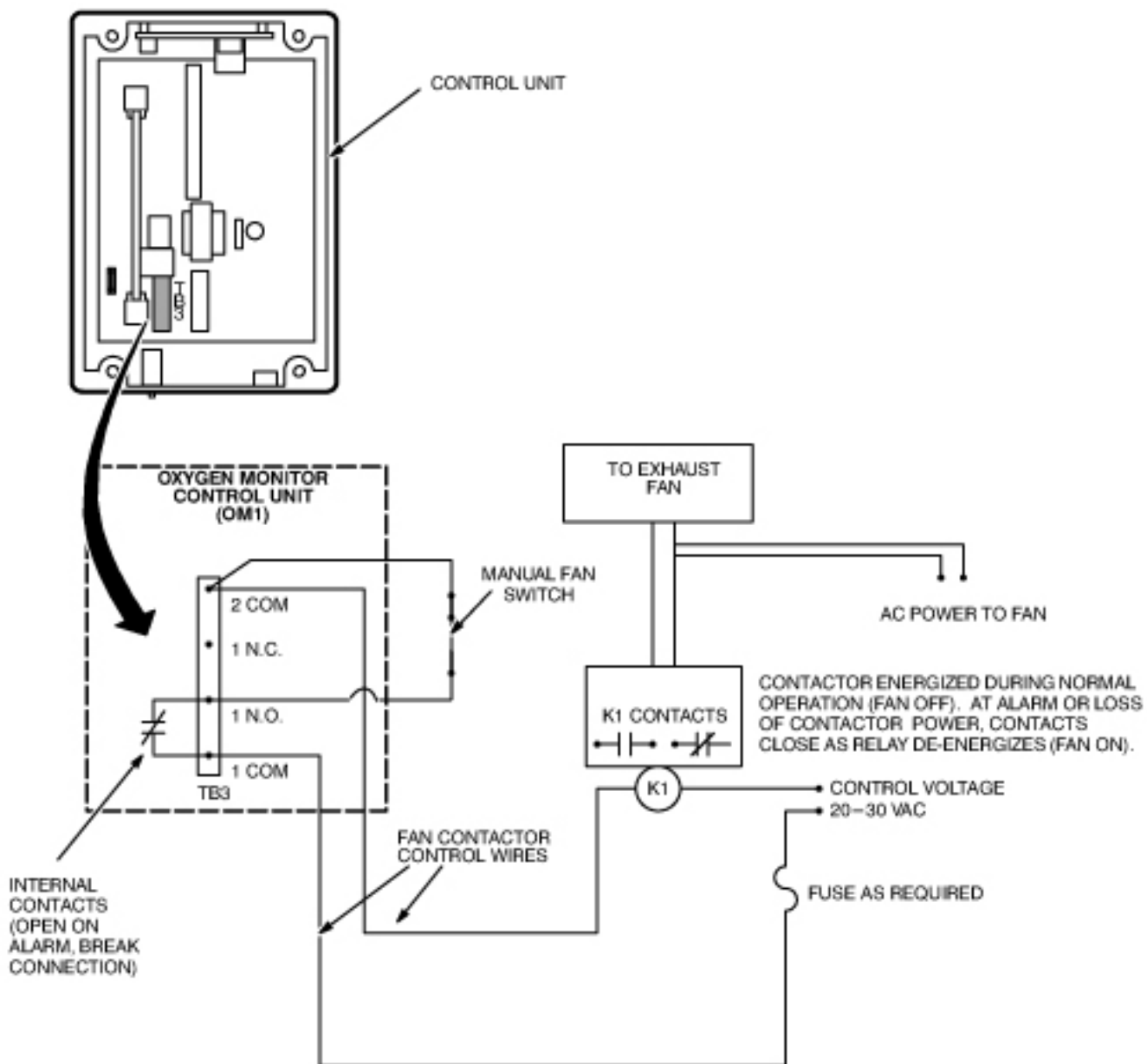


1.1.3 Optional Equipment Installation

A relay-operated exhaust fan circuit can be added to the Oxygen Monitor subsystem as an option. To install the fan circuit, do the following:

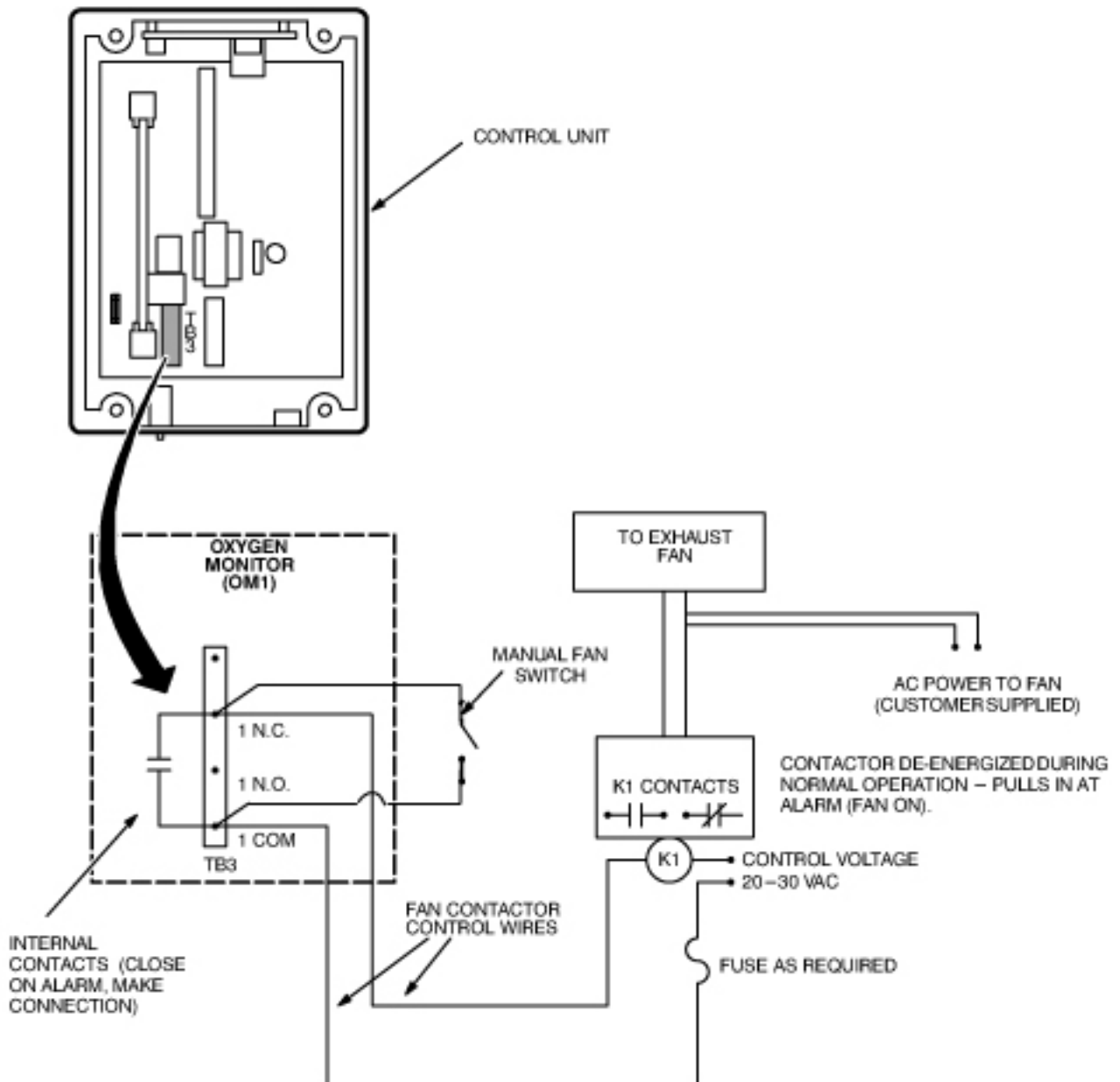
1. Loosen the four captive screws securing the front cover of the Control Unit and open the front cover.
2. Connect the fan circuit to the Control Unit using one of the wiring configurations shown below. Refer to your specific site plan for specific wiring details.

Illustration 4-13: Simplified Fan Circuit Wiring (Normally Energized Fan Contactor)



NOTE: All equipment outside of the control unit (dashed line) is to be supplied by the customer.

Illustration 4-14: Simplified Fan Circuit Wiring (Normally De-Energized Fan Contactor)



NOTE: All equipment outside of the control unit (dashed line) is to be supplied by the customer.

1.1.4 Sensor Retrofit Kit Installation

NOTE: For 5-Year Oxygen Sensors, if you do not have a Sensor Retrofit Plate, you need to obtain and install one. If you do not use a Sensor Retrofit Plate, the 5-Year Oxygen Sensor will not fit properly in the remote sensor module.

New Remote Sensor Modules may have sensors mounted so that an Oxygen Calibration Unit screw-in nozzle can be attached to the sensor during calibration of the Oxygen Monitor Subsystem. A Sensor Retrofit Kit (46-301672P3) is also available to retrofit older Remote Sensor Modules to

allow attachment of the Oxygen Calibration Unit nozzle. To install the Sensor Retrofit Kit, do the following:

1. Remove the front cover of the Remote Sensor Module.
2. Remove the Oxygen Sensor Cell and Sensor Circuit Board from the front cover as shown in [Illustration 4-15](#).
3. With a flat-tip screwdriver, remove the glue securing the porous metal disk to the back of the front cover, and remove the disk as shown in [Illustration 4-16](#).

NOTE: During installation of the Sensor Retrofit Plate, make sure the gasket side of the plate is installed against the sensor cell. Also, due to variations in sensor circuit board design, two sets of screws are provided in the kit. Be sure to use the appropriate length screws so that the sensor cell and circuit board are securely fastened to the Sensor Retrofit Plate.

4. Install the Oxygen Sensor Cell, Sensor Circuit Board, and Sensor Retrofit Plate on the front cover with the two screws provided in the kit. See [Illustration 4-17](#).

Illustration 4-15: Removing the Oxygen Sensor Cell and Sensor Circuit Board

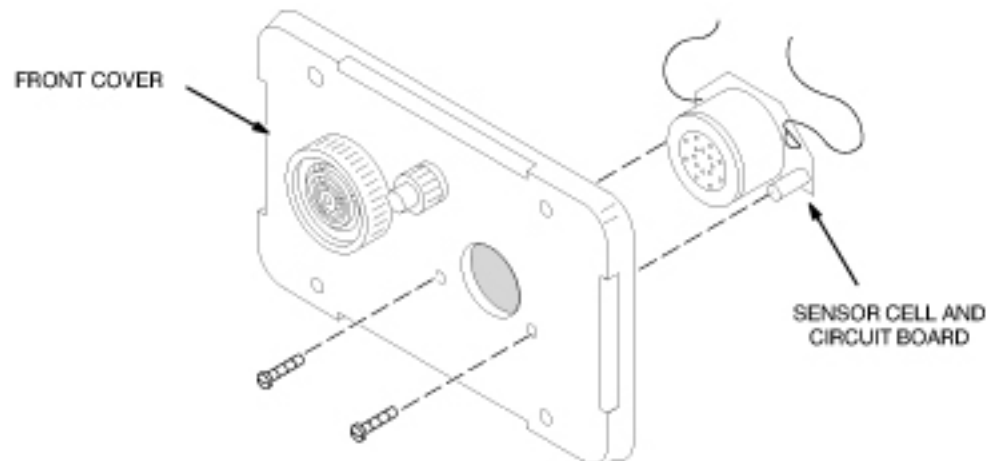


Illustration 4-16: Removing the Porous Metal Disk

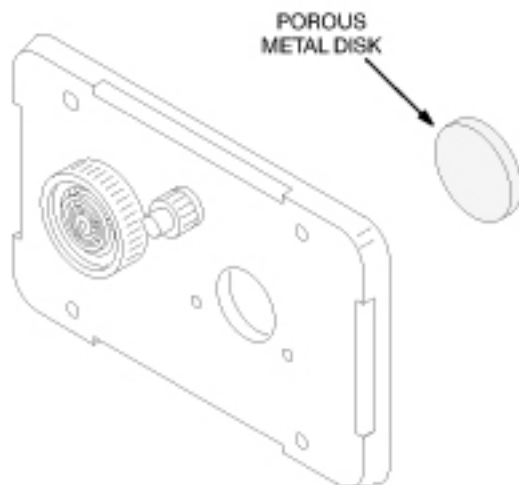
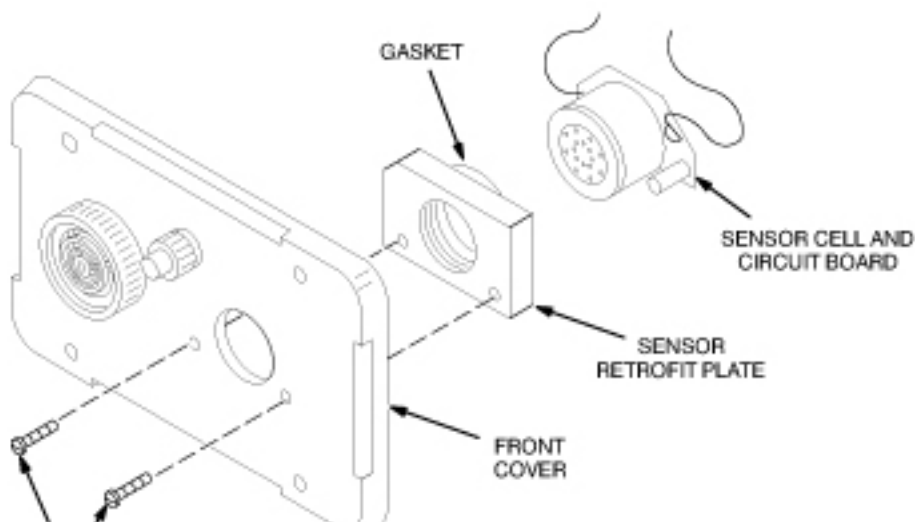


Illustration 4-17: Installing the Sensor Retrofit Kit



1.2 Model ISA-60M

See Vendor Manual, part number 80003-500 for the installation process for the ISA-60M Oxygen Monitor. Additional information is provided below.

To access the Oxygen Monitor internal wiring connections, remove the four screws shown below.

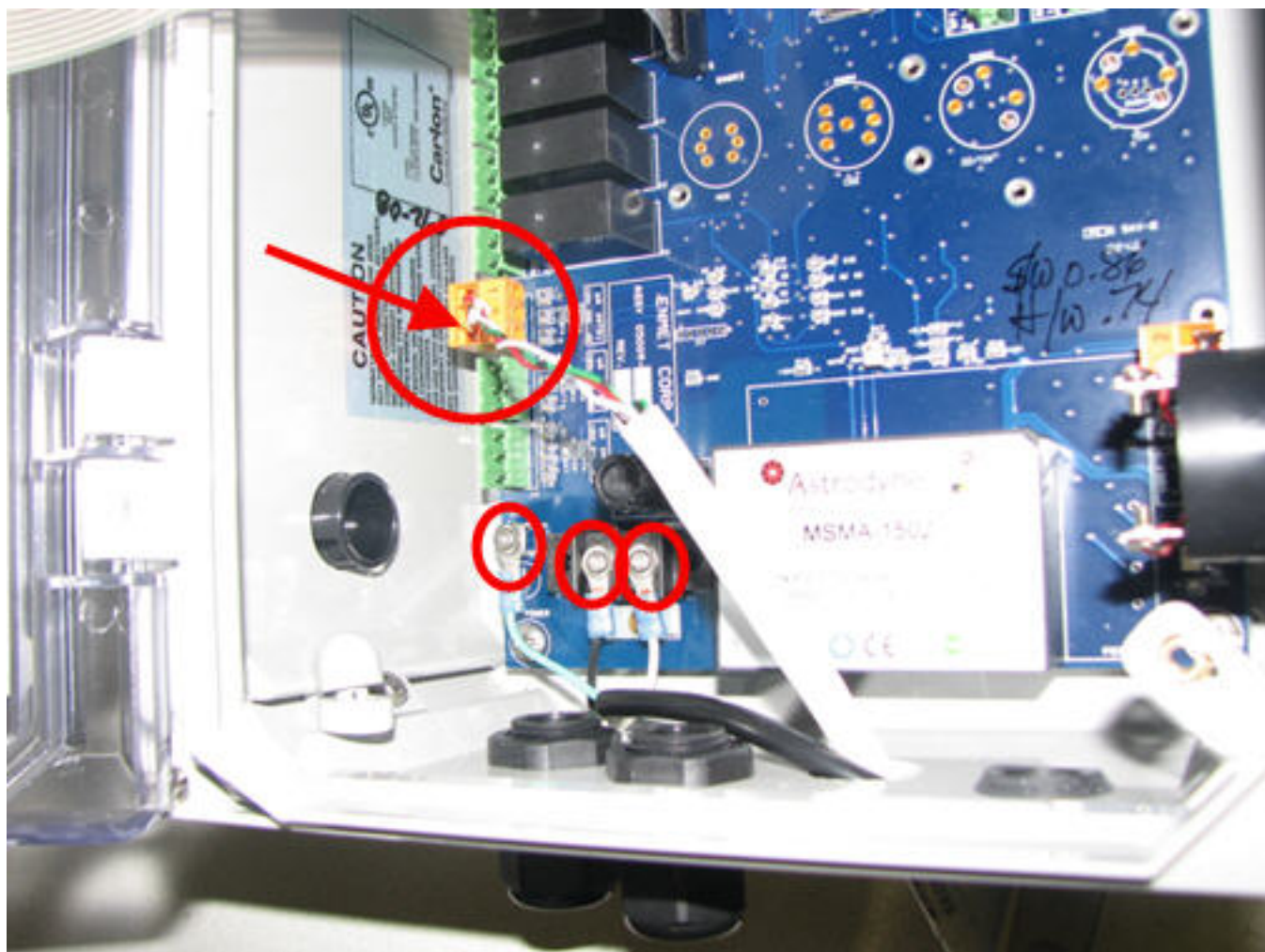
NOTE: Cable passage through the enclosure should utilize strain reliefs whenever possible.

Illustration 4-18: Access Internal Wiring Connections



Connect the remote sensor cable and power connections as shown below.

Illustration 4-19: Connections to O2 Monitor

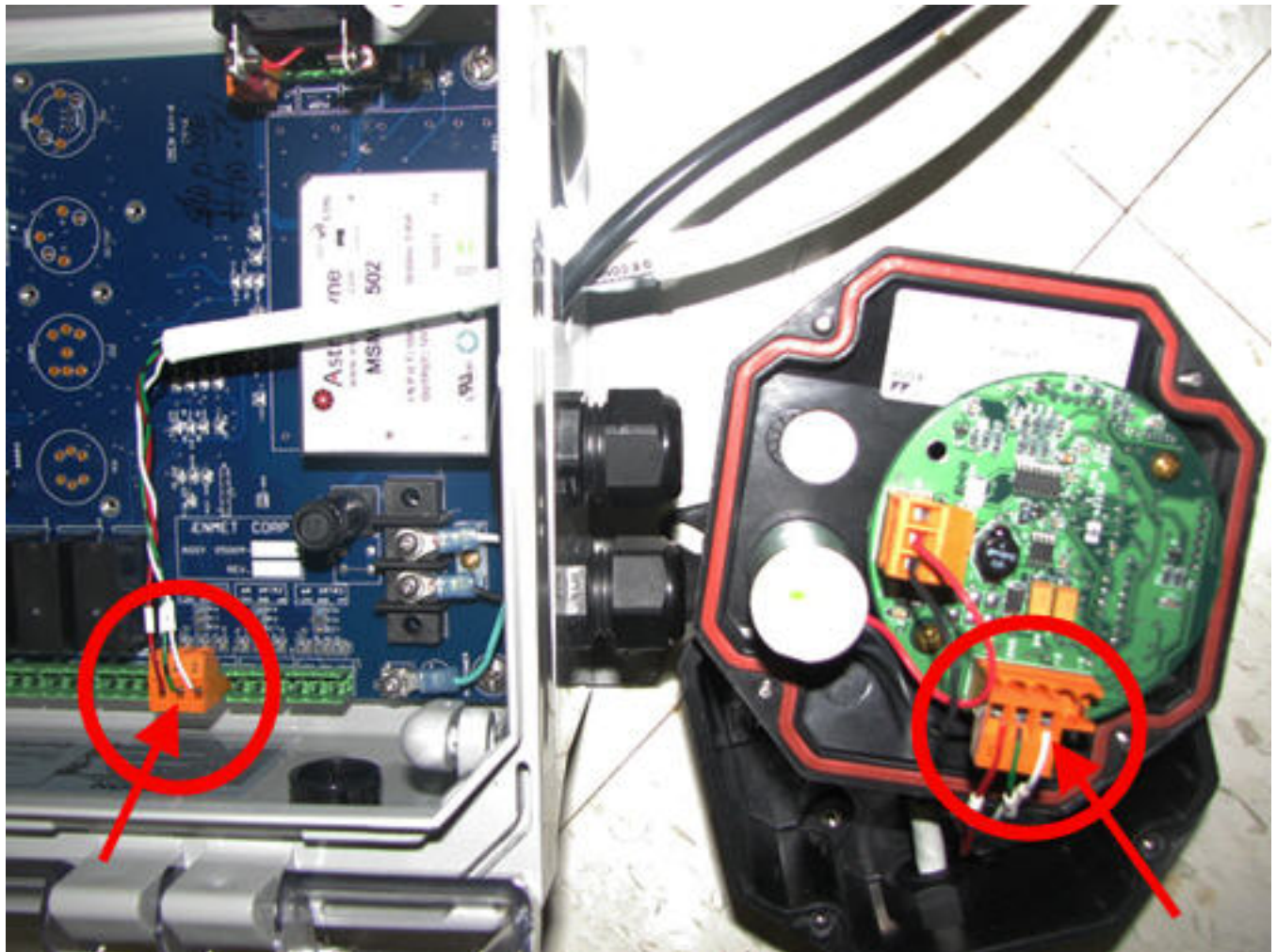


When connecting the cabling between the Oxygen Monitor and Oxygen Sensor, make sure the following connections are maintained between Monitor and Sensor:

- 24V
- Gnd
- MA

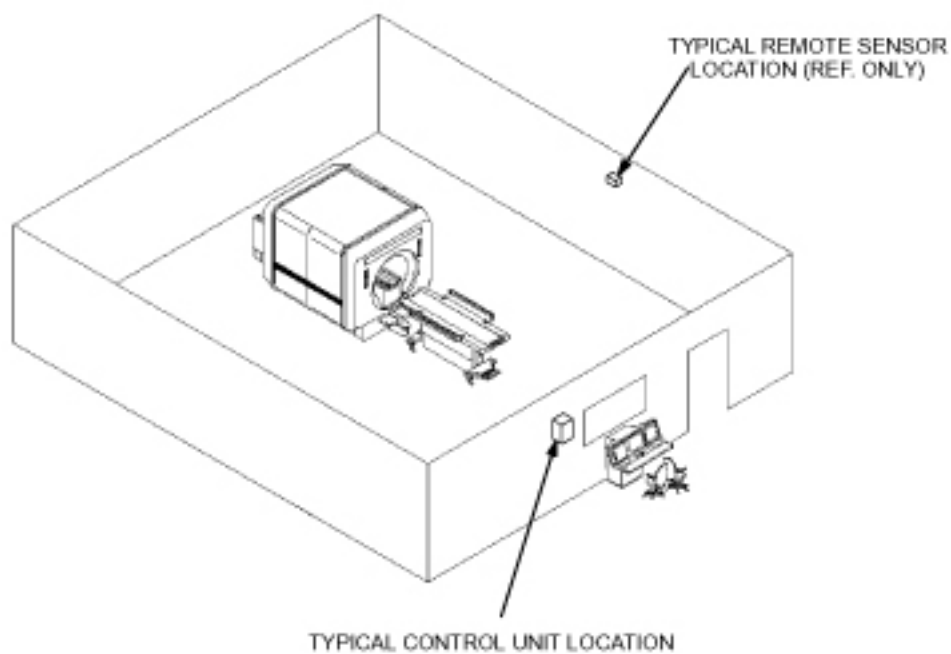
See below.

Illustration 4-20: Ensure Cabling Connections Maintained



See below for a typical Oxygen Monitor Subsystem layout.

Illustration 4-21: Typical Oxygen Monitor Subsystem Installation



See [Illustration 4-12](#) for Oxygen Monitor connections at the Pen Panel.

Chapter 5 Calibration

1 Calibration

1.1 Model ISA-42M

NOTE: See Vendor Manual, part number 80003-500 for the calibration process for the ISA-60M Oxygen Monitor.

There are two calibration procedures for the Oxygen Monitor subsystem. The aerosol gas calibration procedure ([Section 1.1.1](#)) and the oxygen tank/regulator procedure ([Section 1.1.2](#)). To use either procedure, a Sensor Retrofit Kit (46-320936P1) must be installed. See [Illustration 5-1](#). If missing, order and install before continuing.

1.1.1 Aerosol Gas Calibration Procedure

1.1.1.1 Overview

This section contains the procedures for calibrating the Oxygen Monitor subsystem. This is one of two types of calibration that are performed on the Oxygen Monitor subsystem:

- **Gain adjustment.** This adjustment is performed:
 - At initial installation of the Oxygen Monitor subsystem.
 - After replacing the Main Circuit board or Oxygen Sensor Cell.
 - After circuit alignment calibration.
 - According to your site's periodic maintenance schedule.
- **Circuit alignment.** The Oxygen Monitor subsystem is shipped pre-aligned but may require realignment if the alarm does not operate properly or when the 5-year cell is installed.

NOTE: This section describes the calibration procedures using the newer Aerosol Gas Calibration Kit (2173689). The Remote Sensor Module must contain a Sensor Retrofit Kit (46-320936P1). See [Illustration 5-1](#). If missing, order and install before continuing.

1.1.1.2 Required Tools

The tools required for calibrating the Oxygen Monitor are:

- Oxygen Calibration Kit: GE P/N 2173689 (CAC-2554)
- Small screwdriver
- Voltmeter (only used for circuit alignment)

This procedure has been updated to simplify the calibration process. The gas cylinder calibration kit (46-328021G1) has been replaced by Aerosol Gas Calibration Kit (5399042). This kit includes two aerosol containers, one filled with 20.9% oxygen, balance nitrogen, and one filled with 17.0% oxygen, balance nitrogen. These aerosol containers are constructed of aluminum and can be

safely brought into the MRI room. For the purposes of calibration/test, the aerosol containers provide approximately 3 individual calibrations. After use, these cans can be discarded.

NOTE: Gas flowing time is limited to 30 - 45 seconds per can. These aerosol cans can be used for up to three calibrations if they are removed after a 15 second purge attachment. System test should be performed immediately after aerosol removal. It is important to mark the aerosol label where indicated after each test so that filled containers may be distinguished from empty or almost empty containers. For an extended one time use, the aerosol container may be left in place continuously until all gas has been exhausted. Continuous discharge offers the highest accuracy over the time gas is available in the aerosol container.

NOTE: In order to conserve gas, it is necessary to trap a known volume of a gas mixture between the calibration adapter and the Oxygen Sensor. To properly accomplish this, an O-ring has been included with this kit which must be permanently installed in the Oxygen Sensor port. The O-ring should be placed by hand or with a needle nose pliers at the bottom of the threaded retrofit block firmly against the foam ring of the Oxygen Sensor. Refer to [Illustration 5-1](#) for placement details.

1.1.1.3 Gain Adjustment and Alarm Functional Check

1. Turn on the power to the Control Unit. The green power LED should light.
2. Screw the Aerosol Gas Calibration Adapter (2173691) into the gas sensing port (which contains the retrofit block) located on the front cover of the Remote Sensor Module. See [Illustration 5-1](#).
3. Using the aerosol can marked 20.9% Oxygen found in the Oxygen Calibration Kit, remove the cap and insert the top of the can with the nozzle inserted into the Aerosol Calibration Adapter. It will be necessary to snap the can onto the plastic fingers of the adapter by first attaching the can at an angle. Apply moderate force until the aerosol can snaps into place. When properly attached, you will be able to hear the gas automatically discharging. See [Illustration 5-2](#), [Illustration 5-3](#), and [Illustration 5-4](#).

NOTE: If a new Oxygen Sensor cell was installed, wait at least one hour for the sensor to stabilize, then readjust the oxygen gain until the meter reads 20.9%.

4. Wait approximately 15 seconds (After 15 seconds, aerosol container may be removed for repeated tests or left in place for continuous gas discharge). Immediately adjust the O₂ GAIN adjustment potentiometer on the Control Unit until the meter reads 20.9% oxygen ([Illustration 5-5](#)). Immediately remove the aerosol can by tilting the can at an angle until the container snaps free from the calibration adapter. Retain for possible repeated tests.
5. Using the aerosol can marked 17.0% Oxygen found in the Oxygen Calibration Kit, remove the cap and insert the top of the can with the nozzle directed into the Aerosol Calibration Adapter. It may be necessary to snap the can onto the plastic fingers of the adapter by first attaching the can at an angle. Apply moderate force until the aerosol can snaps into place. When properly attached, you will be able to hear the gas automatically discharging. See [Illustration 5-2](#), [Illustration 5-3](#), and [Illustration 5-4](#).
6. With the 17% oxygen aerosol attached to the Remote Sensor Module, the following should occur after approximately 15 seconds. (See [Illustration 5-5](#)):

- The meter reading on the Control Unit should decrease to less than 18%.
 - The audible alarm should sound on the Control Unit and Remote Sensor Module.
 - The red alarm LED should blink on the Control Unit and Remote Sensor Module.
7. Immediately disconnect the 17.0% aerosol container from the Remote Sensor Module (after approximately 15 seconds). Disconnect the Aerosol Calibration Adapter to allow the ambient air with normal oxygen content to contact the sensor. Retain the 17% aerosol container for possible repeated tests.
- NOTE:** Unscrewing the Aerosol Calibration Adapter from the Remote Sensor Module allows rapid diffusion of ambient oxygen to reach the sensor cell. Unscrew adaptor only if aerosol container is removed to prevent unnecessary gas waste.
8. Press the **Relay Release** button on the front cover of the Control Unit. The following should occur:
- The meter on the Control Unit should read approximately 20.9%.
 - The audible alarms should stop sounding.
 - The red alarm LEDs should stop blinking and stay off.
9. If the system responds correctly as per Steps 1 through 8, your calibration is complete. Otherwise, continue with [Section 1.1.1.4](#), Circuit Alignment.

Illustration 5-1: Control Unit Meter and Gain Adjustment Potentiometer

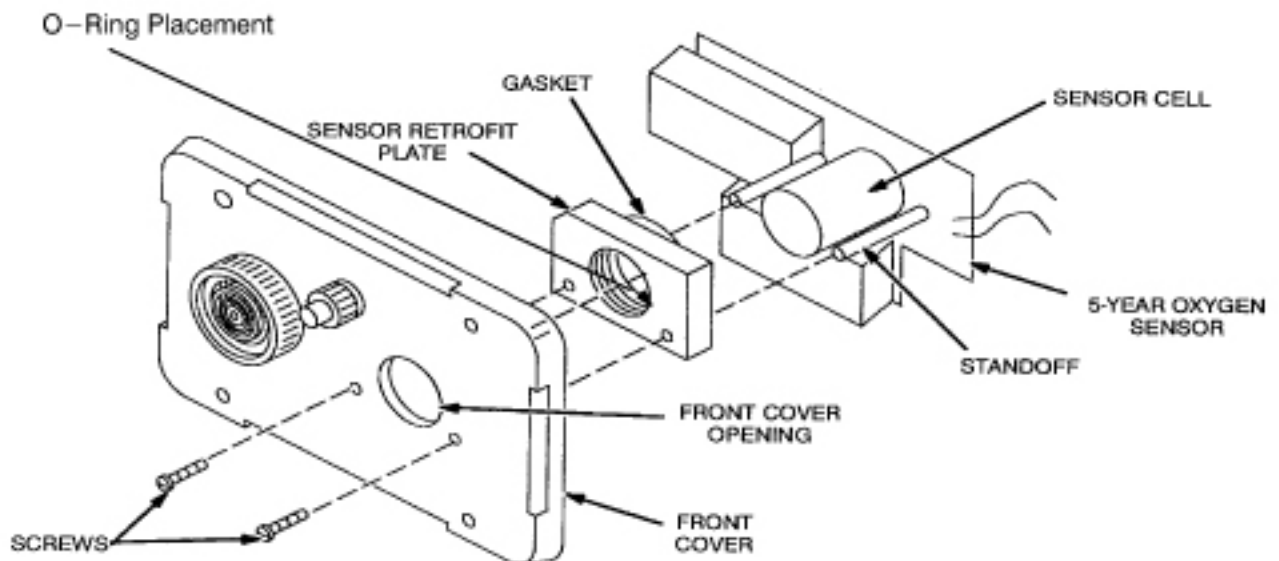


Illustration 5-2: Attaching Calibration Tool to the Oxygen Sensor Port

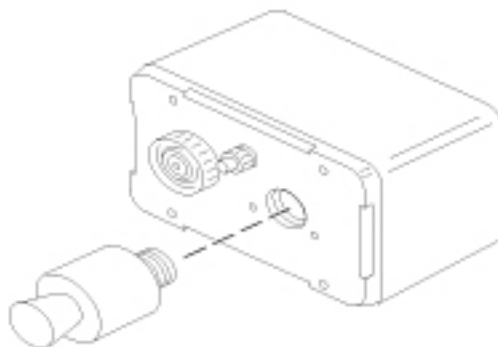


Illustration 5-3: Insert Oxygen Can Into the Calibration Tool

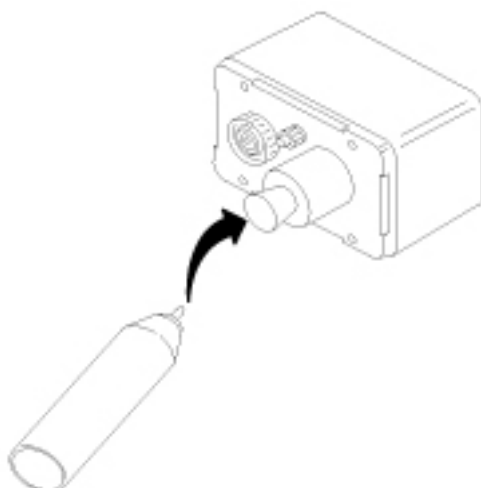


Illustration 5-4: Apply Moderate Force Until Can Snaps Into Place

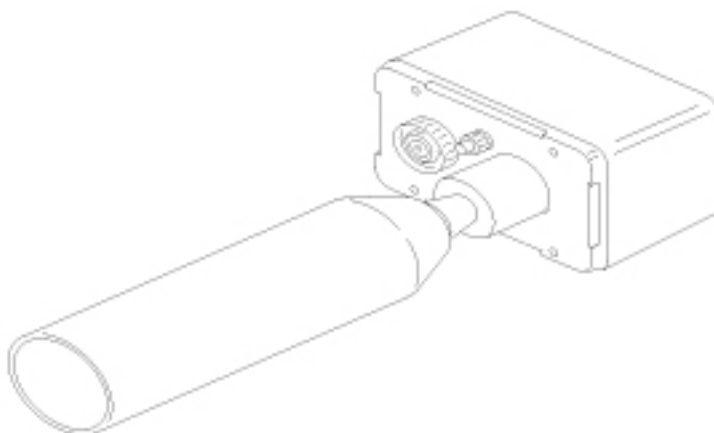
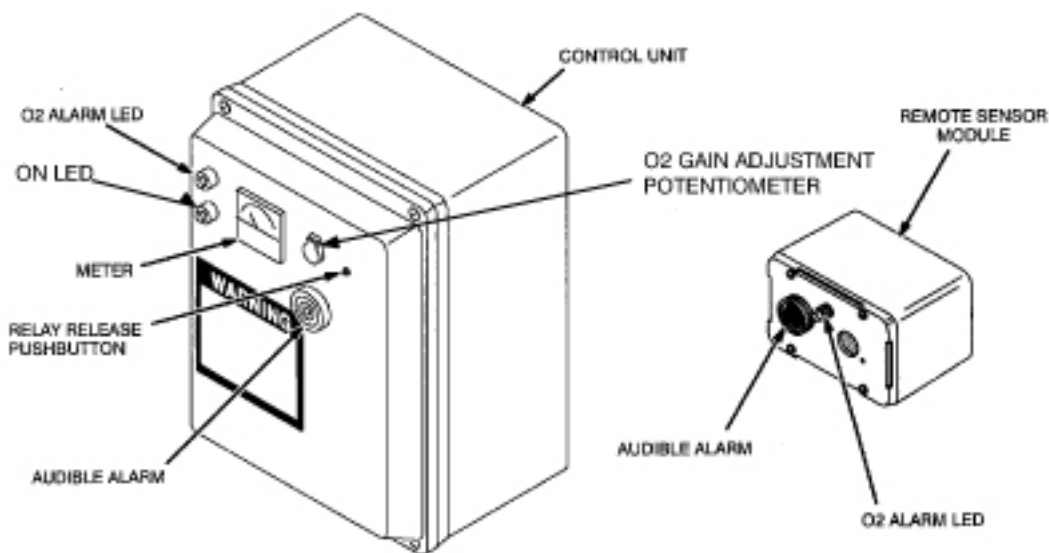


Illustration 5-5: Control Unit Meter and Gain Adjustment Potentiometer

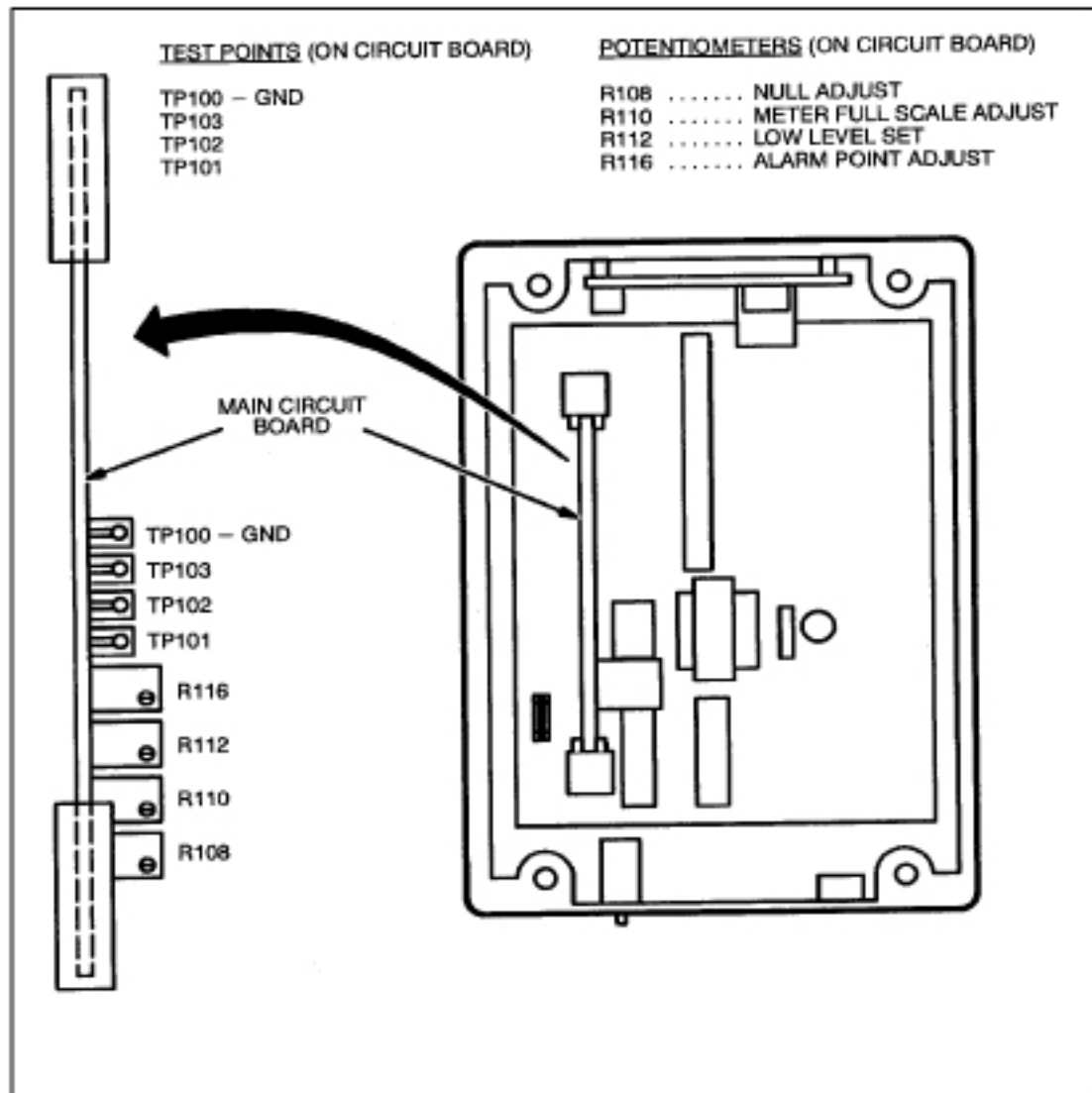
**NOTICE**

Because of the annoying and possibly alarming sound made by the Sonalert, it may be disabled during testing by disconnecting its red wire. Be sure to reconnect the wire after testing! Attaching a 10K resistor in series with the Sonalert permits audible alarm operation at a reduced volume.

1.1.1.4 Circuit Alignment

The location of the Main Circuit Board and the test points and potentiometers used during the circuit alignment procedures are shown below.

Illustration 5-6: Test Point And Potentiometer Locations

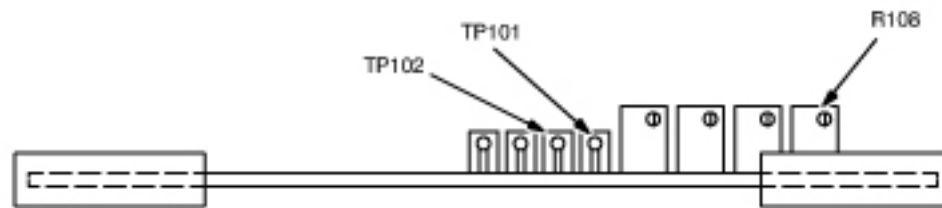


Null Adjustment

Null adjustment is not required when using the 5-year cell. Due to differences between this cell and the 18-month cell, it is only necessary to verify that the null adjustment is set fully counterclockwise.

Adjust the null potentiometer R108 fully clockwise (at least 10 turns) and then fully counterclockwise (at least 10 turns) or until a click is felt or heard indicating end of travel has been reached. This sweep adjustment is intended to clean the trim potentiometer wiper of contamination which might lead to an intermittent contact condition. See [Illustration 5-7](#). During this verification, audible and visual alarm should activate. Press **Relay Release** button to restart system.

Illustration 5-7: Null Adjustment

**Low Level Set Adjustment**

1. Measure the voltage between the Oxygen Sensor circuit board lead wires. These leads are the two wires leading from the 5-year cell. Connection may also be made inside the ISA-40 Control Unit where the Oxygen Sensor cable enters the box. This is typically on the bottom across barrier strip connections 4 and 5. (Lead wires should be color coded Orange and Green. Connect your meter ground to the Green and meter positive to Orange. See [Illustration 5-8](#).

The voltage should read between 0.030 and 0.060 VDC inclusive.

NOTE: If voltage across Oxygen Sensor lead wires is below 0.030 VDC (or until 20.9% oxygen can not be achieved with the O₂ gain control tuned fully clockwise (CW)) , replace the Oxygen Sensor GE P/N 2112207. (Remember: Voltage readings for Oxygen Sensors must be made when sensor is attached to ISA-40 oxygen monitor circuit, otherwise, for an isolated 5-year cell, a 2.2 K resistor must be first connected across sensor leads. Voltage measurement can now be made across leads.)

2. Connect the negative lead of the voltmeter to test point TP101 and the positive lead to test point TP103 on the Control Unit main circuit board. See [Illustration 5-9](#).
3. Adjust Low Level Set potentiometer R112 until the voltmeter reads 1.03 VDC.

Illustration 5-8: Alarm Adjustment

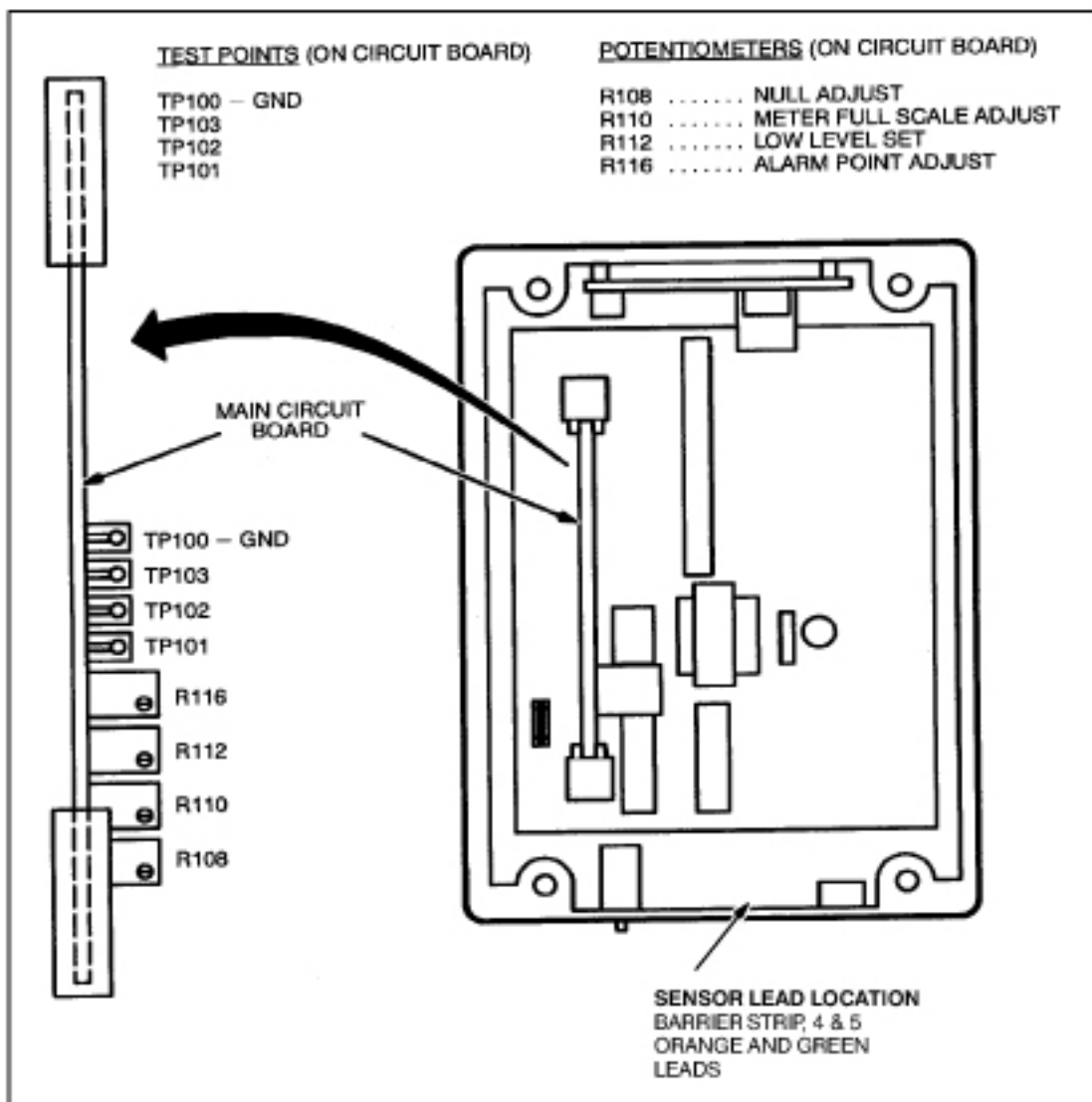
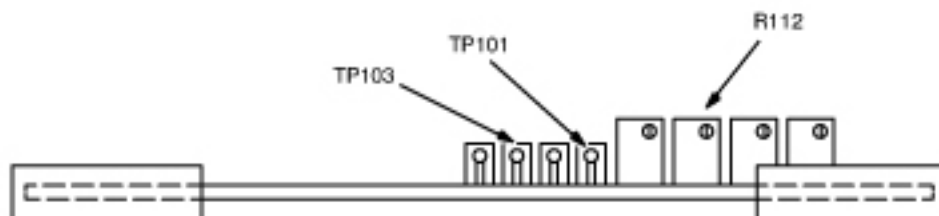


Illustration 5-9: Low Level Set Adjustment

**Meter Full Scale Adjustment**

1. Connect the negative lead of the voltmeter to test point TP101 and the positive lead to test point TP102 on the Control Unit main circuit board. See [Illustration 5-10](#).

2. Adjust the O₂ GAIN adjustment potentiometer on the front cover of the Control Unit until the voltmeter reads 1.55 VDC. See [Illustration 5-11](#). If 1.55 VDC is not attainable, set adjustment @ full CW position.
3. Adjust the Meter Full Scale Adjustment potentiometer R110 until the meter on the front cover of the Control Unit reads full scale (26%). Counterclockwise increases the meter reading, Clockwise decreases the meter reading. See [Illustration 5-10](#).
4. Remove the voltmeter from the test points.

Illustration 5-10: Meter Full Scale Adjustment

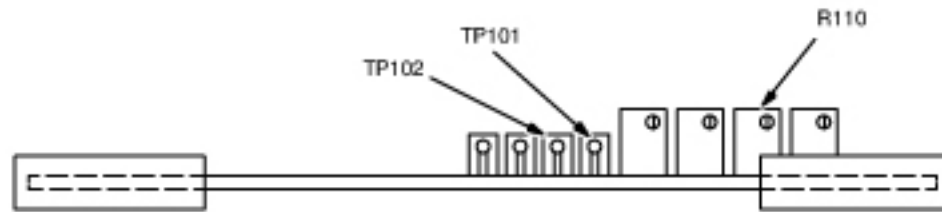
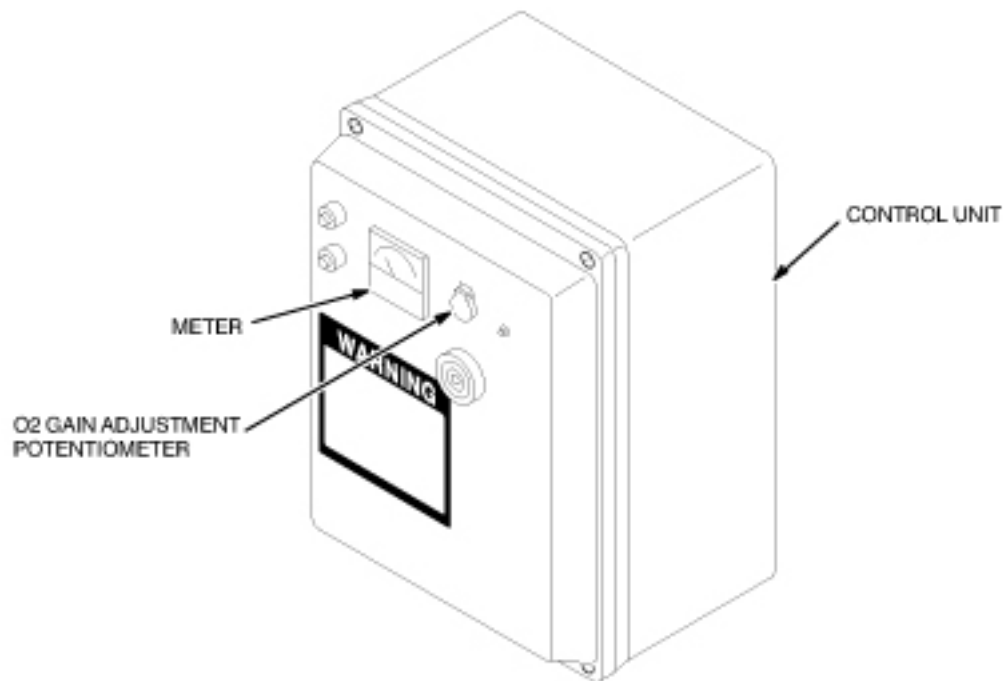


Illustration 5-11: Control Unit Meter and Gain Adjustment Potentiometer

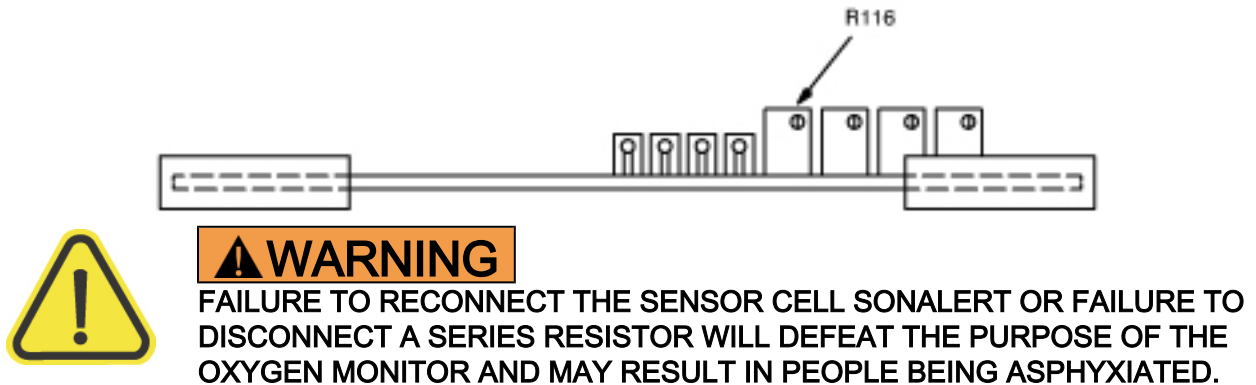


Alarm Set Adjustment

1. Adjust the O₂ GAIN adjustment potentiometer on the front cover of the Control Unit until the meter reads 18.0%. If the red alarm LEDs on the Control Unit and Remote Sensor Module start blinking before the meter reads 18.0%, the Alarm Set Point is set too high. To lower the set point:
 - a. Rotate the Alarm Point Adjust potentiometer R116 clockwise a few turns, see [Illustration 5-12](#).

- b. Press the **Relay Release** button on the front cover of the Control Unit. The LEDs should remain off. Repeat Steps a and b if the LEDs continue to blink.
 2. Rotate the Alarm potentiometer R116 counterclockwise until the Control Unit Alarm begins to activate. The following should occur:
 - The audible alarm should sound on the Control Unit and Remote Sensor Module.
 - The red alarm LEDs should blink on the Control Unit and Remote Sensor Module.
 3. Adjust the O₂ GAIN adjustment potentiometer on the front cover until the meter reads greater than 18.0% and press the **Relay Release** button. The alarm LEDs should stop blinking and remain off.
 4. Adjust the O₂ GAIN adjustment potentiometer until the alarm LEDs blink. If the LEDs do not start blinking when the meter reads 18.0%, repeat Steps 2 through 4. Otherwise, continue with Step 5.
 5. Adjust the O₂ GAIN adjustment potentiometer until the meter reads 20.9%, then push the **Relay Reset** button. Verify 20.9% reading using aerosol adapter and calibration aerosol gas per [Section 1.1.1.3](#). Adjust O₂ GAIN if necessary.
 6. Reattach the aerosol adapter to the Oxygen Sensor port and attach the 17% oxygen aerosol to the Remote Sensor Module. Wait at least 15 seconds. The following should occur:
 - The meter reading on the Control Unit should decrease to less than 18%.
 - The audible alarms should sound on the Control Unit and Remote Sensor Module.
 - The red alarm LEDs should blink on the Control Unit and Remote Sensor Module.
- NOTE:** If the low oxygen alarm does not activate properly, replace the 5-Year Oxygen Sensor cell assembly, GE P/N 2112207, then obtain another 2173689 calibration kit and repeat [Section 1.1.1.3](#).
7. Remove the aerosol container and the aerosol adapter. Wait approximately 30 seconds for ambient air to diffuse into oxygen sensing port. Press the **Relay Release** button.
 8. The following should occur:
 - The meter on the Control Unit should read approximately 20.9%.
 - The audible alarms should stop sounding.
 - The red alarm LEDs should stop blinking and stay off.
 9. Reconnect the red wire to the Sonalert, if previously disconnected. Calibration and circuit alignment is now complete.

Illustration 5-12: Alarm Adjustment



1.1.2 Oxygen Tank/Regulator Calibration Procedure

1.1.2.1 Overview

- **Gain adjustment.** This adjustment is performed:
 - At initial installation of the Oxygen Monitor subsystem.
 - After replacing the Main Circuit board or Oxygen Sensor Cell.
 - After circuit alignment calibration.
 - According to your site's periodic maintenance schedule.
- **Circuit alignment.** The Oxygen Monitor subsystem is shipped pre-aligned but may require realignment if the alarm does not operate properly.

NOTE: This section describes the calibration procedures using the newer Oxygen Calibration Kit (46-328021G1), and it assumes that the Sensor Retrofit Kit (46-320936P1) has been installed on the Remote Sensor Module. Make adjustments to the procedure as necessary, if the equipment conditions are different than documented in this section.

1.1.2.2 Required Tools

The tools required for calibrating the Oxygen Monitor are:

- Oxygen Calibration Kit (46-317271G1).
- Small screwdriver.
- Voltmeter (only used for circuit alignment).

1.1.2.3 Gain Adjustment and Alarm Functional Check

1. Turn on the power to the Control Unit. The green power LED should light.
2. Screw the hose barb from the Oxygen Calibration Kit onto the front cover of the Remote Sensor Module. See [Illustration 5-13](#).

3. Connect one end of the hose from the calibration kit to the 20.9% regulator hose barb. Leave the other end of the hose unattached at this time, so that the hose can be purged when the gas is turned on.
4. With the 20.9% regulator pressure adjust valve closed (turned completely counterclockwise) and the regulator toggle valve closed, open the 20.9% gas cylinder valve. The inlet pressure gauge will show the cylinder pressure.

NOTE: Adjust the valves slowly and only use low pressures. Adjusting the valves too fast or adjusting to excessive pressures may damage the sensor cell.

5. Open the 20.9% regulator toggle valve and slowly adjust the regulator pressure adjust valve for an outlet pressure of one or two PSIG as indicated on the outlet pressure gauge.
6. Connect the other end of the hose to the hose barb on the Remote Sensor Module.
7. Adjust the O₂ GAIN adjustment potentiometer on the Control Unit until the meter reads 20.9% oxygen. See [Illustration 5-14](#).

NOTE: If a new Oxygen Sensor cell was installed, close the 20.9% gas cylinder valve and wait at least four hours for the sensor to stabilize, then readjust the O₂ gain until the meter reads 20.9%.

8. Shut off the 20.9% regulator and gas cylinder valves and remove the hose from the 20.9% regulator hose barb and also from the hose barb on the Remote Sensor Module.
9. Connect one end of the hose to the 17.0% regulator hose barb. Leave the other end of the hose unattached at this time, so that the hose can be purged when the gas is turned on.
10. With the 17.0% regulator pressure adjust valve closed (turned completely counterclockwise) and the regulator toggle valve closed, open the 17.0% gas cylinder valve. The inlet pressure gauge will show the cylinder pressure.
11. Open the 17.0% regulator toggle valve and slowly adjust the regulator pressure adjust valve for an outlet pressure of one or two PSIG as indicated on the outlet pressure gauge.
12. Connect the other end of the hose to the hose barb on the Remote Sensor Module. The following should occur [Illustration 5-15](#):
 - The meter reading on the Control Unit should decrease to less than 18%.
 - The audible alarm should sound on the Control Unit and the Remote Sensor Module.
 - The red alarm LED should blink on the Control Unit and the Remote Sensor Module.
13. Shut off the 17.0% regulator and cylinder valves and remove the hose from the hose barb on the Remote Sensor Module to allow the ambient air with normal oxygen content to contact the sensor.
14. Press the **Relay Release** button on the front cover of the Control Unit. The following should occur:
 - The meter on the Control Unit should read approximately 20.9%.
 - The audible alarms should stop sounding.

- The red alarm LEDs should stop blinking and stay off.

Illustration 5-13: Oxygen Calibration Kit

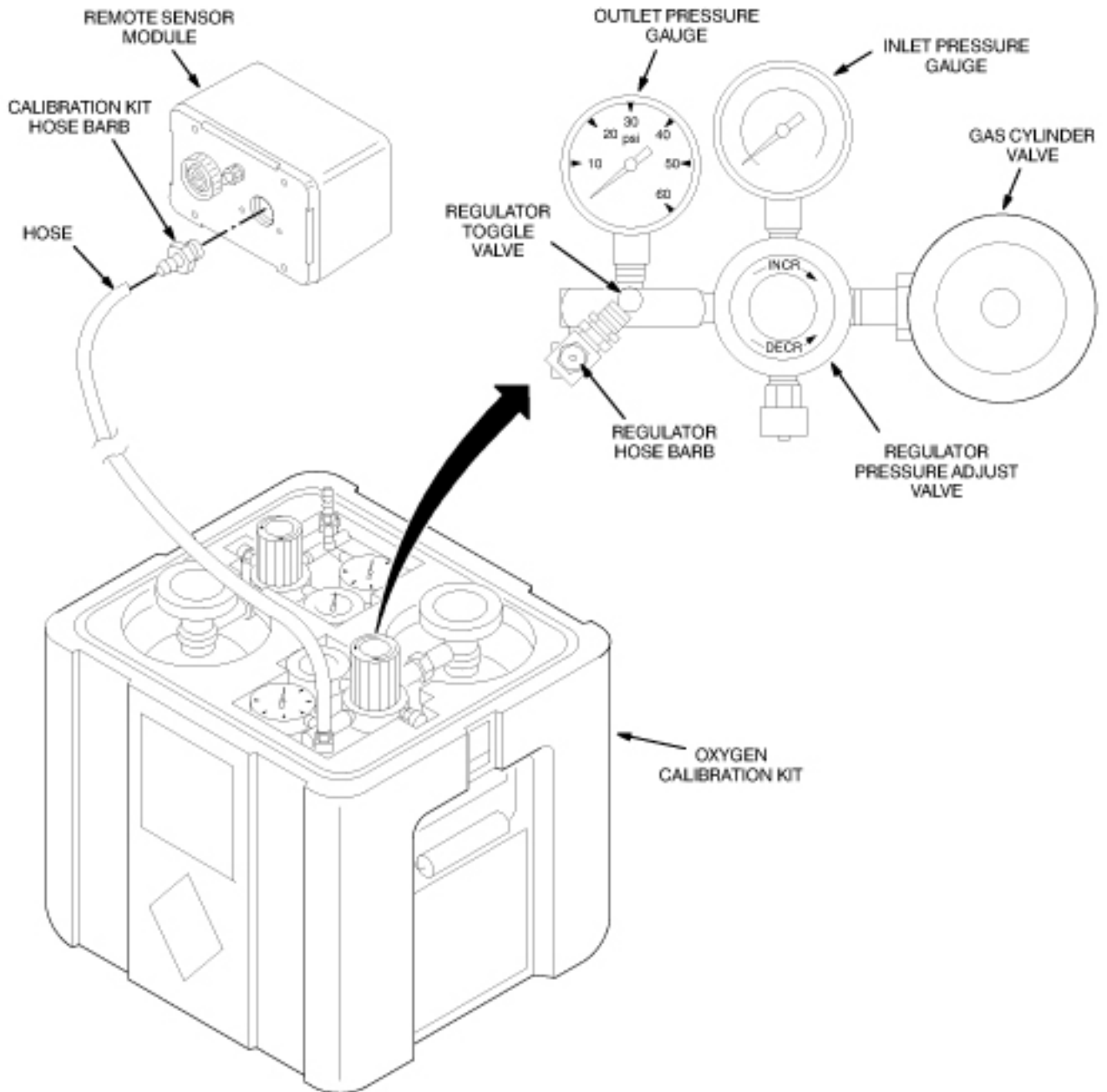


Illustration 5-14: Control Unit Meter And Gain Adjustment Potentiometer

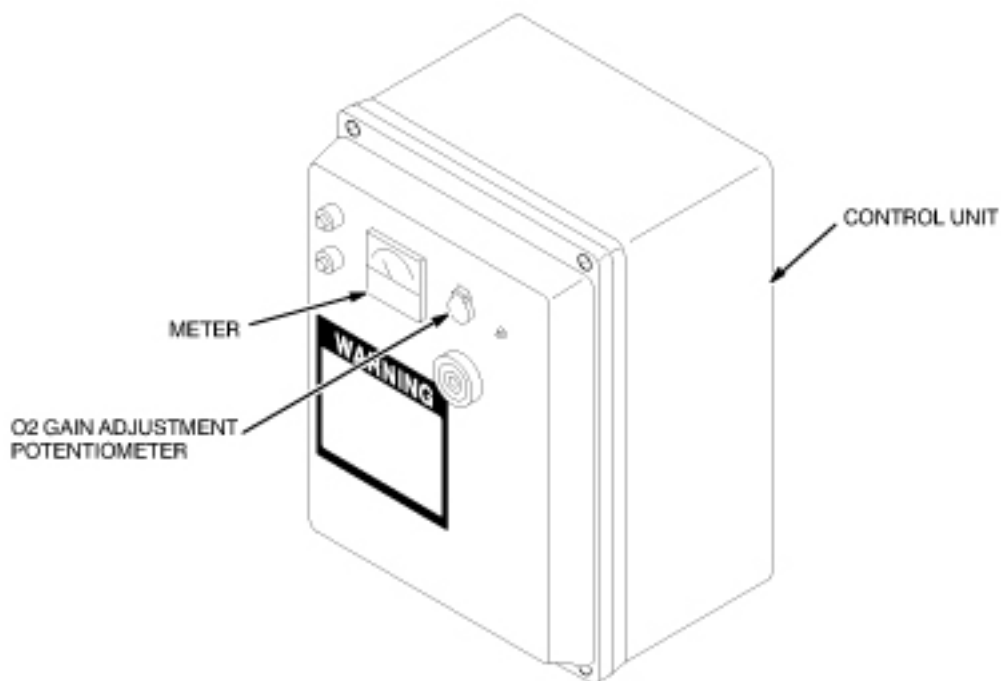
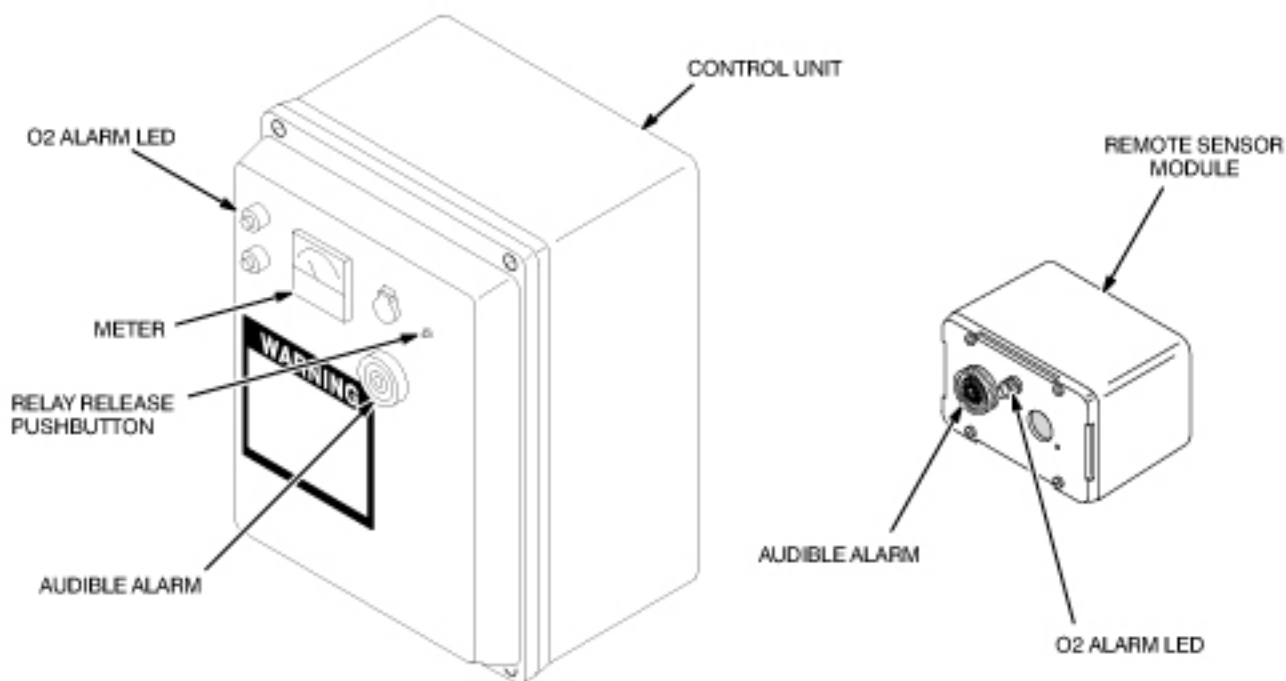


Illustration 5-15: Oxygen Monitor Subsystem Alarms and Relay Release Button

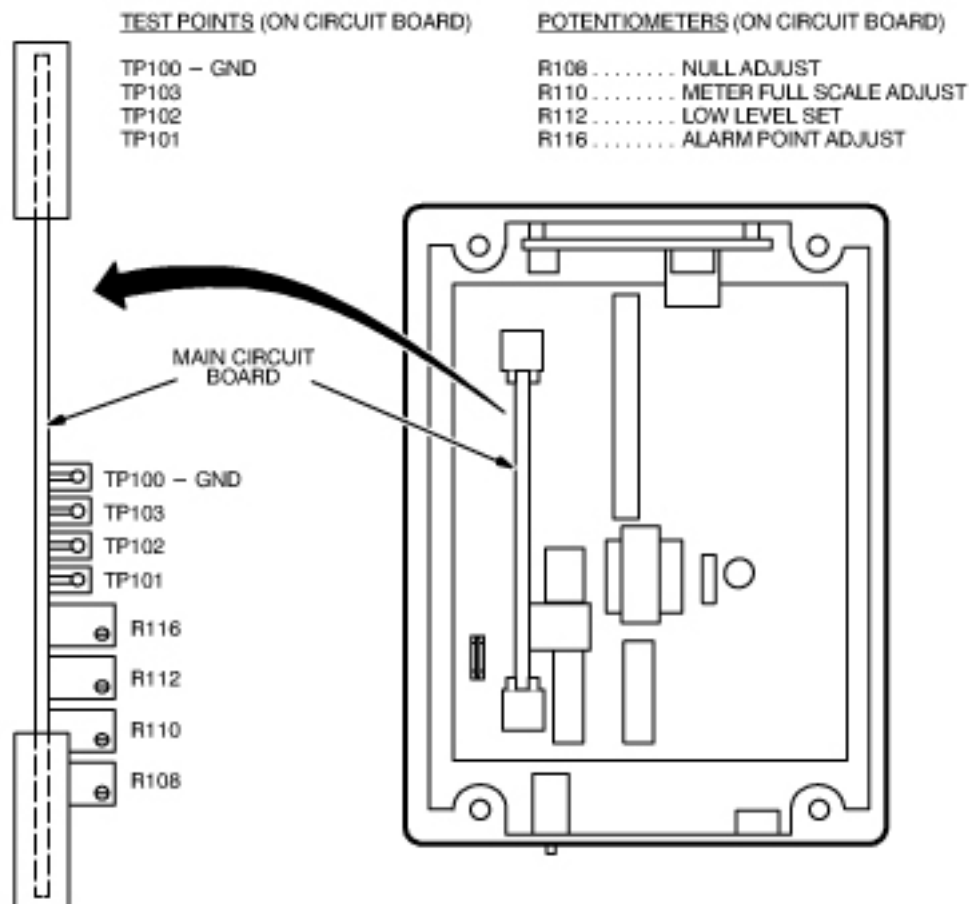


1.1.2.4 Circuit Alignment

NOTE: Because of the annoying and possibly alarming sound made by the Sonalert during testing, disconnect the red wire to the Sonalert during testing and reconnect the wire after testing.

NOTE: The location of the Main Circuit Board and the test points and potentiometers used during the circuit alignment procedures are shown in [Illustration 5-16](#).

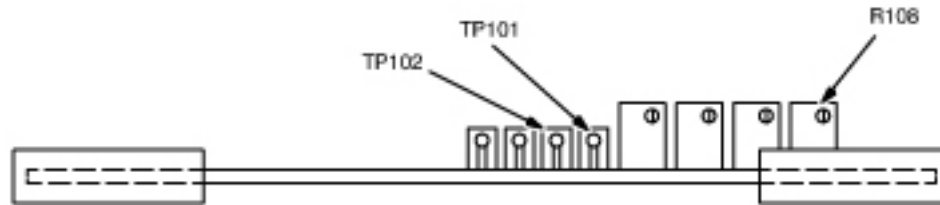
Illustration 5-16: Test Point And Potentiometer Locations



Null Adjustment

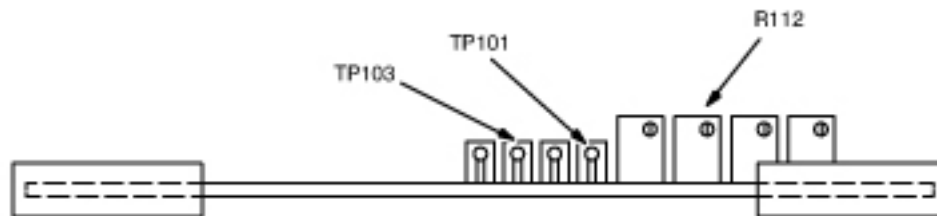
1. Disconnect one of the Oxygen Sensor circuit board wires from its terminal in the Remote Sensor Module.
2. Connect the negative lead of the voltmeter to test point TP101 and the positive lead to test point TP102 on the Control Unit main circuit board. See [Illustration 5-17](#).
3. Adjust Null Adjustment potentiometer R108 until the voltmeter reads 0.00 VDC.
4. Reconnect the Oxygen Sensor circuit board wire to its terminal in the Remote Sensor Module.

Illustration 5-17: Null Adjustment

**Low Level Set Adjustment**

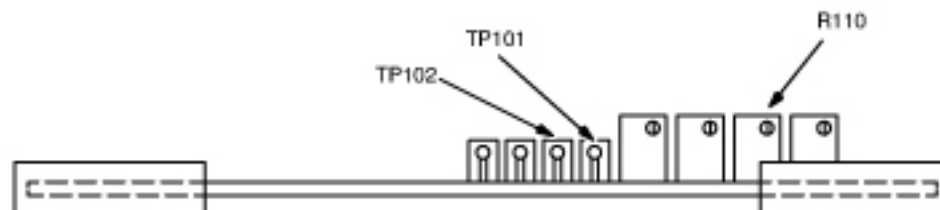
1. Measure the voltage between the Oxygen Sensor circuit board lead wires. The voltage should read between 0.04 and 0.06 VDC
2. Connect the negative lead of the voltmeter to test point TP101 and the positive lead to test point TP103 on the Control Unit main circuit board. See below.
3. Adjust Low Level Set potentiometer R112 until the voltmeter reads 1.03 VDC.

Illustration 5-18: Low Level Set Adjustment

**Meter Full Scale Adjustment**

1. Connect the negative lead of the voltmeter to test point TP101 and the positive lead to test point TP102 on the Control Unit main circuit board.
2. Adjust the O₂ GAIN adjustment potentiometer on the front cover of the Control Unit until the voltmeter reads 1.55 VDC. See [Illustration 5-14](#).
3. Adjust the Meter Full Scale Adjustment potentiometer R110 until the meter on the front cover of the Control Unit reads full scale (26%). Clockwise increases the meter reading, counterclockwise decreases the meter reading. See [Illustration 5-19](#).
4. Remove the voltmeter from the test points.

Illustration 5-19: Meter Full Scale Adjustment

**Alarm Set Adjustment**

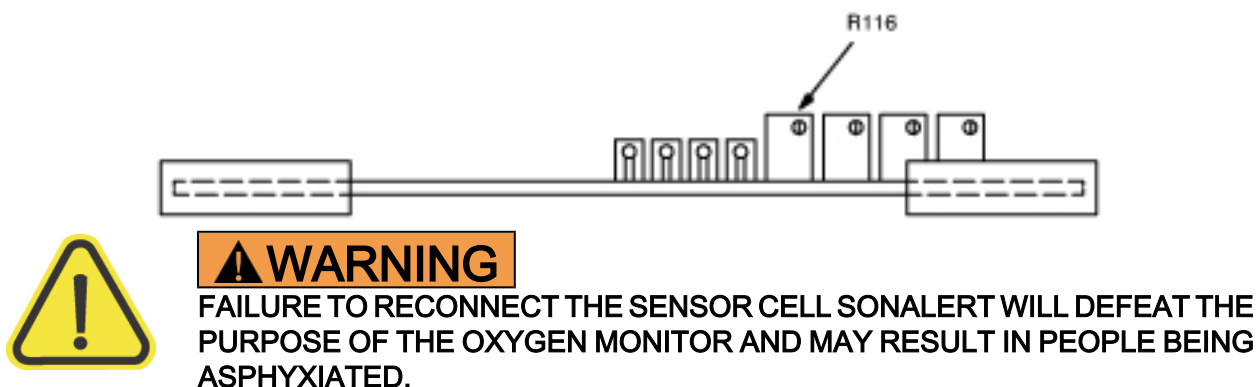
1. Adjust the potentiometer on the front cover of the Control Unit until the meter reads 18.0%. If the red alarm LEDs on the Control Unit and Remote Sensor Module start blinking before the meter reads 18.0%, the Alarm Set Point is set too high. To lower the set point:
 - a. Rotate the Alarm Point Adjust potentiometer R116 clockwise a few turns. See [Illustration 5-20](#)
 - b. Press the **Relay Release** button on the front cover of the Control Unit. The LEDs should remain off. Repeat Steps a and b if the LEDs continue to blink.
2. Rotate the Alarm Point Adjust potentiometer R116 counterclockwise until the Control Unit Alarm begins to activate. The following should occur:
 - The audible alarm should sound on the Control Unit and Remote Sensor Module.
 - The red alarm LEDs should blink on the Control Unit and Remote Sensor Module.
3. Adjust the O₂ GAIN adjustment potentiometer until the meter reads greater than 18.0% and press the **Relay Release** button. The alarm LEDs should stop blinking and remain off.
4. Adjust the O₂ GAIN adjustment potentiometer until the alarm LEDs blink. If the LEDs do not start blinking when the meter reads 18.0%, repeat Steps 2 through 4. Otherwise, continue with Step 5.
5. Adjust the O₂ GAIN adjustment potentiometer until the meter reads 20.9%, then push the **Relay Reset** button.
6. Screw the hose barb from the Oxygen Calibration Kit onto the front cover of the Remote Sensor Module. See [Illustration 5-14](#).
7. Connect one end of the hose to the 17.0% regulator hose barb. Leave the other end of the hose unattached at this time, so that the hose can be purged when the gas is turned on.
8. With the 17.0% regulator pressure adjust valve closed (turned completely counterclockwise) and the regulator toggle valve closed, open the 17.0% gas cylinder valve. The inlet pressure gauge will show the cylinder pressure.

NOTE: Open the valves slowly and only use low pressures. Opening the valve too fast or adjusting to excessive pressures may damage the sensor cell.
9. Open the 17.0% regulator toggle valve and slowly adjust the regulator pressure adjust valve for an outlet pressure of one or two PSIG as indicated on the outlet pressure gauge.
10. Connect the other end of the hose to the hose barb on the Remote Sensor Module. The following should occur:
 - The meter reading on the Control Unit should decrease to less than 18%.
 - The audible alarms should sound on the Control Unit and Remote Sensor Module.
 - The red alarm LEDs should blink on the Control Unit and Remote Sensor Module.
11. If the low oxygen alarm does not activate properly, repeat the entire Calibration section. If the Oxygen Monitor system does not operate properly after readjustment, replace the Oxygen

Sensor cell and circuit board refer to [Chapter 6, Replacement](#) for replacement procedures. Then repeat [Calibration](#).

12. Press the **Relay Release** button.
13. Reconnect the red wire to the Sonalert, if previously disconnected.

Illustration 5-20: Alarm Set Adjustment



1.2 Model ISA-60M

See Vendor Manual, part number 80003-500 for the calibration process for the ISA-60M Oxygen Monitor.

Chapter 6 Replacement

1 Replacement

1.1 Model ISA-42M

NOTE: See Vendor Manual, part number 80003-500 for replacement processes for the ISA-60M Oxygen Monitor.

1.1.1 Fuse

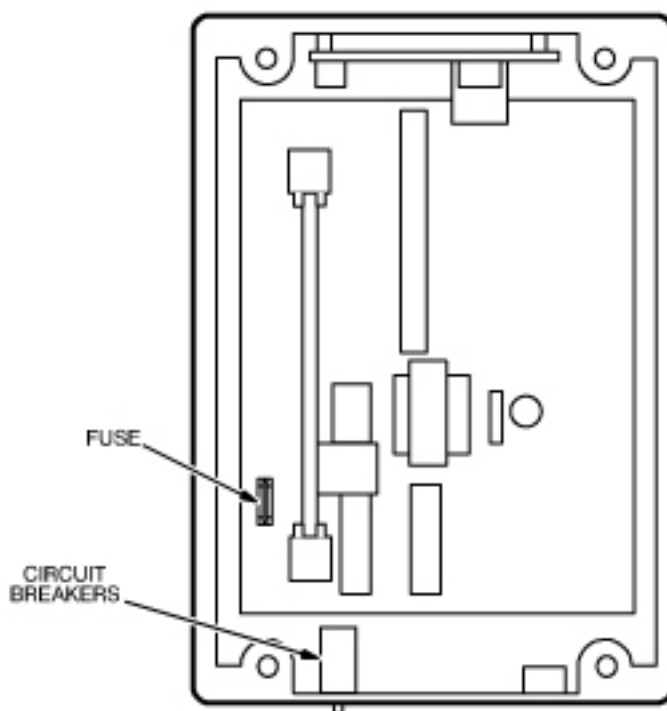
The Control Unit contains a 50 mA slow blow fuse. If all LEDs are off with external power being supplied to the Control Unit, the fuse may have blown.

1. Before replacing the fuse, do the following:
 - a. Press the circuit breaker reset buttons on the bottom of the Control Unit to verify that the circuit breakers are not tripped.
 - b. If the Control Unit power LED does not come on, turn off external power to the Control Unit.
2. Open the front cover on the Control Unit and remove the fuse. See [Illustration 6-1](#).

NOTE: Fuse replacement requires exact replacement with the same type of fuse. Any other type of fuse may result in damage to the Control Unit.

3. Verify that the fuse no longer has continuity. Replace the fuse if it does not have electrical continuity.

Illustration 6-1: Fuse And Circuit Breaker Locations



1.1.2 LEDs

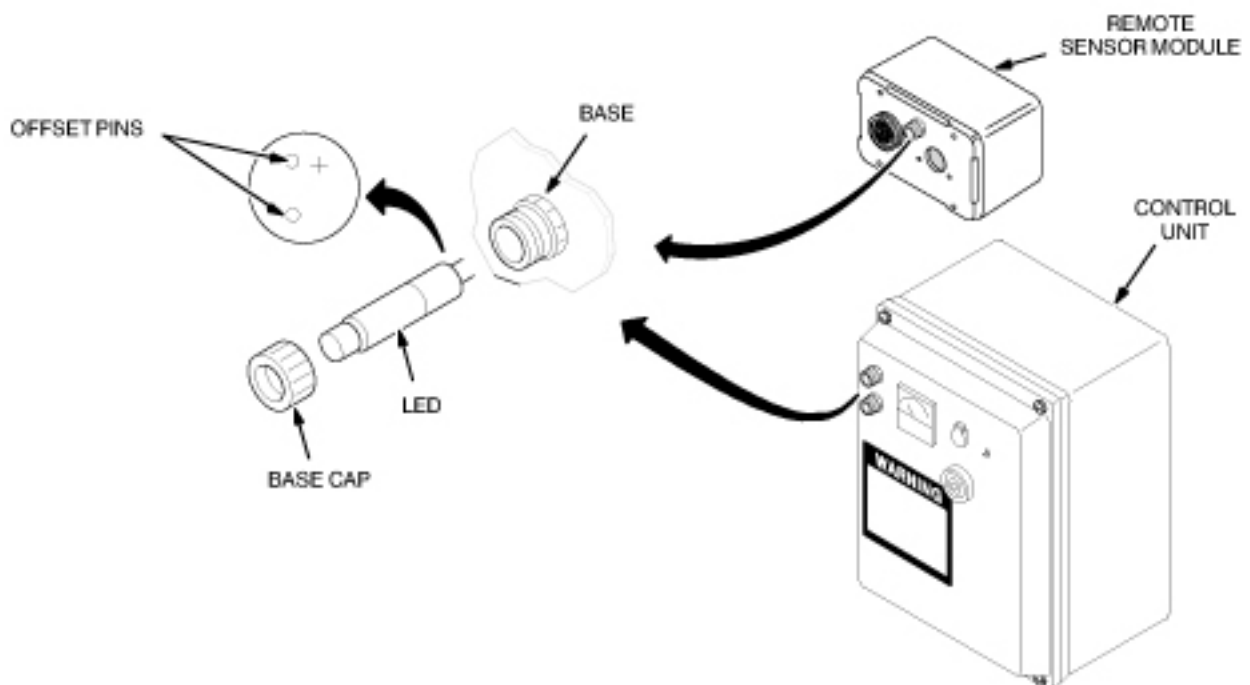
The Control Unit contains a green power LED and red alarm LED. The Remote Sensor Module has a red alarm LED. To replace the LEDs:

1. Turn off the external power to the Control Unit.
2. Unscrew the LED base cap and pull out the LED as shown in [Illustration 6-2](#).
3. Align the offset pins of the new LED with the holes in the base on the Remote Sensor Module or Control Unit.

NOTE: If the LED does not fit into the base easily, do not force the LED into the base. The pin offset is very slight and the LED could be forced into the base with the incorrect polarity. The LED will not operate if it is installed with the incorrect polarity.

4. Push in the new LED and screw on the LED base cap.
5. Turn on the external power.

Illustration 6-2: Removing LEDs



1.1.3 Horn

To replace the horn on the Control Unit or Remote Sensor Module:

1. Turn off external power to the Control Unit.
2. Loosen four captive screws to open the front cover on the Control Unit or remove the front cover of the Remote Sensor Module. See [Illustration 6-3](#).
3. Disconnect the wires from the horn.

4. Unscrew the ring securing the horn to the front cover and remove the horn. See [Illustration 6-4](#).
5. Insert the new horn and secure it to the front panel.
6. Reconnect the red wire to the positive horn terminal and the black wire to the negative horn terminal. See [Illustration 6-3](#).
7. Reinstall the front cover and turn on the external power.

Illustration 6-3: Removing Horn Wires

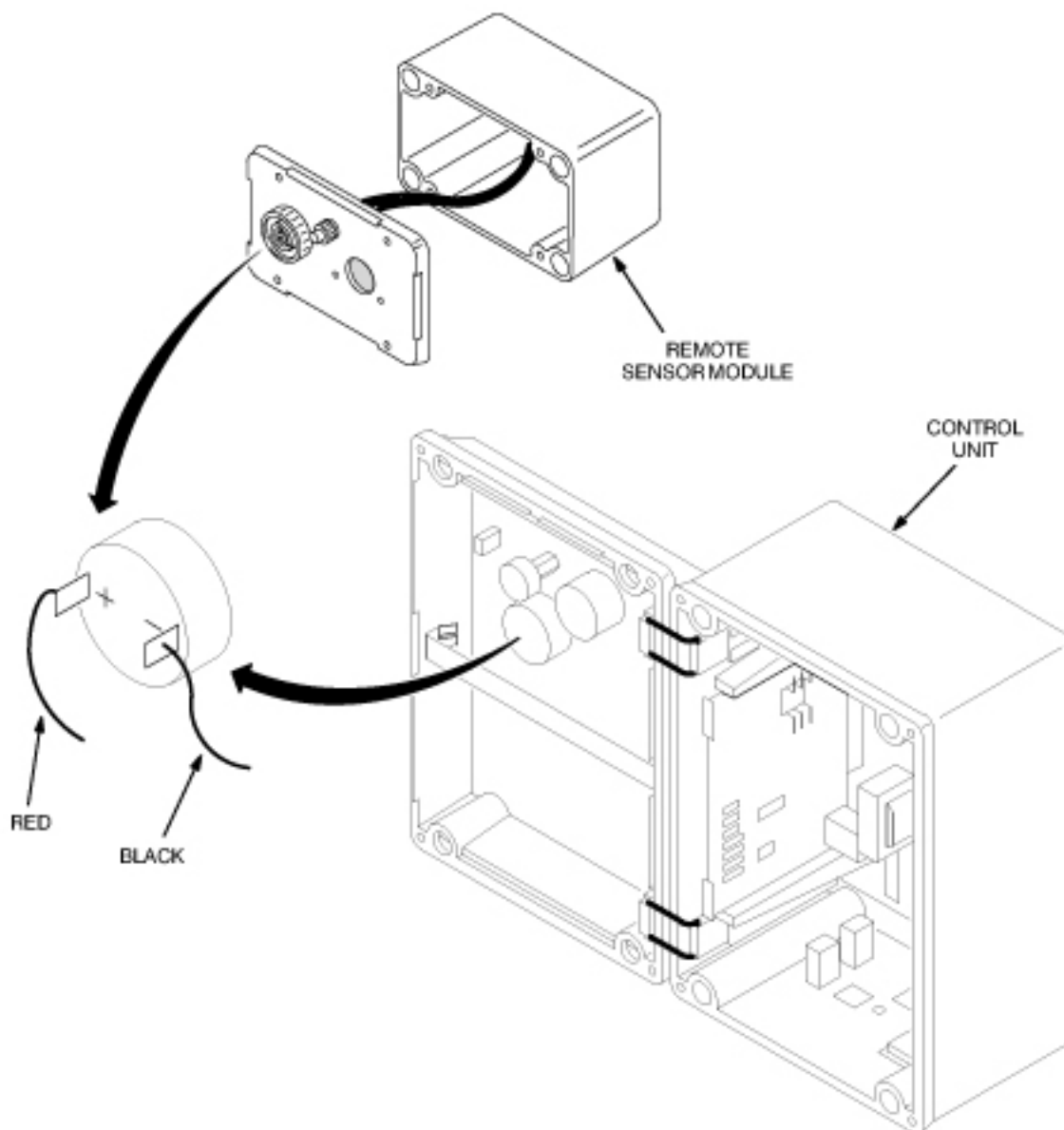
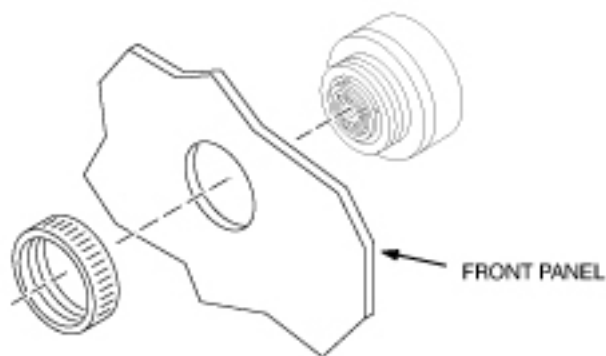


Illustration 6-4: Removing Horn

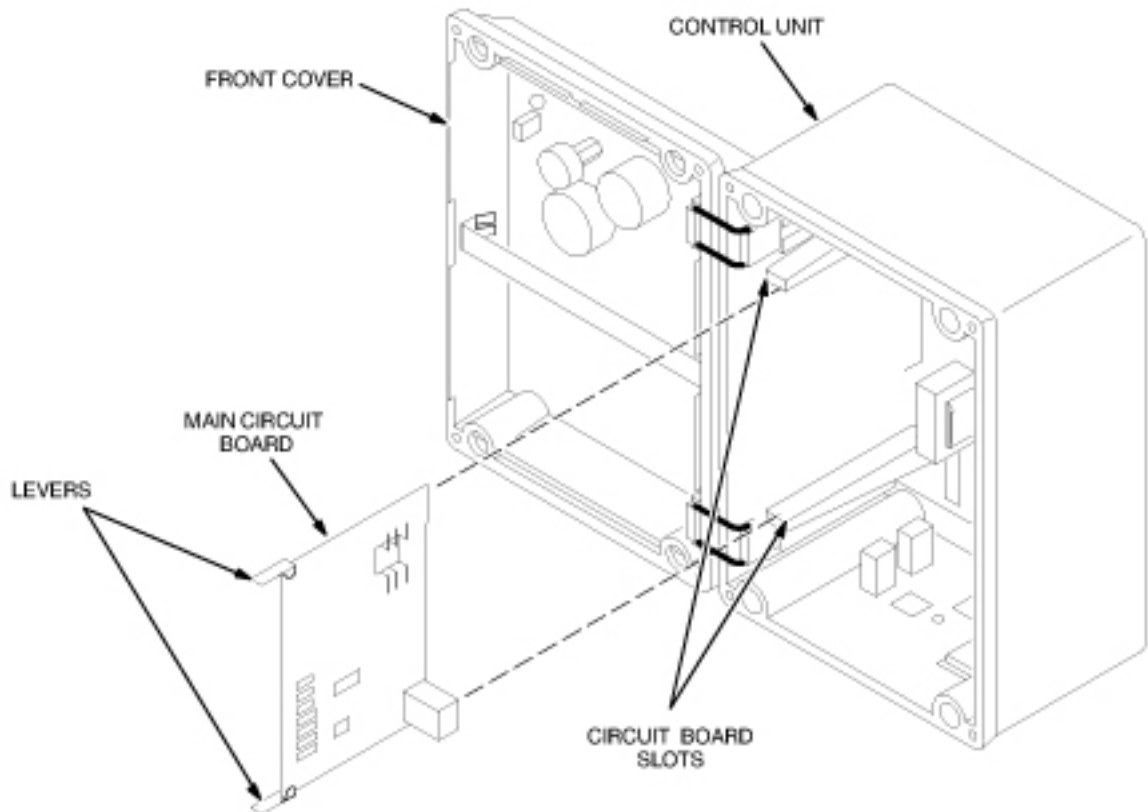


1.1.4 Main Circuit Board

To replace the Main Circuit Board:

1. Turn off external power to the Control Unit.
2. Loosen four captive screws securing the front cover of the Control Unit and open the front cover. See [Illustration 6-5](#).
3. Lift the two levers on the top and bottom of the Main Circuit Board and pull the board out of the Control Unit.
4. Carefully slide the new Main Circuit Board into the circuit board slots in the Control Unit.
5. Close the front cover and turn on external power.
6. Perform the following calibration procedures:
 - a. Circuit Alignment for 5-Year or 18-Month oxygen sensor
 - b. Gain Adjustment and Alarm Functional Check for 5-Year or 18-Month oxygen sensor

Illustration 6-5: Removing The Main Circuit Board



1.1.5 Oxygen Sensor Cell (5-Year or 18-Month)



WARNING

BURN HAZARD! IF THE OXYGEN SENSOR CELL IS LEAKING OR PUNCTURED FOR ANY REASON, PLACE THE CELL ASIDE AND WASH HANDS AND ANY OTHER PARTS OF BODY OR CLOTHING THAT MIGHT HAVE COME IN CONTACT WITH THE ELECTROLYTE LIQUID (KOH SOLUTION). USE LARGE QUANTITIES OF WATER; THE ELECTROLYTE IS A STRONG CAUSTIC SOLUTION AND CAN CAUSE SEVERE BURNS.



NOTICE

If electrolyte liquid or crystal growth are visible anywhere on the cell or within the barrier bag, the cell is leaking and should not be used.

To replace the Oxygen Sensor Cell:

1. Turn off external power to the Remote Sensor Module.
2. Loosen four captive screws and remove the front cover of the Remote Sensor Module. See [Illustration 6-6](#).

3. Remove the two screws securing the Sensor Retrofit Plate, Oxygen Sensor Cell, and Sensor Circuit Board to the front cover.
4. Remove the Oxygen Sensor Cell from the Oxygen Sensor Circuit Board and discard the faulty cell per local regulations
5. Remove the new Oxygen Sensor Cell from the environmental gas barrier bag and remove the shorting clip. See [Illustration 6-7](#).
6. Install the new cell into the Oxygen Sensor Circuit Board making sure the cell center pin is inserted into the center socket of the Oxygen Sensor circuit board. See [Illustration 6-6](#).

NOTE: During installation of the Sensor Retrofit Plate, make sure the gasket side of the plate is installed against the sensor cell.

7. Install and secure the Sensor Retrofit Plate, Oxygen Sensor Cell, and Sensor Circuit Board to the front cover of the Remote Sensor Module with two screws.
8. Install and secure the front cover of the Remote Sensor Module and turn on external power.
9. Perform procedure Gain Adjustment and Alarm Functional Check for 5-Year Oxygen Sensor or Gain Adjustment and Alarm Functional Check for 18-Month Oxygen Sensor.

Illustration 6-6: Removing The Oxygen Sensor Cell

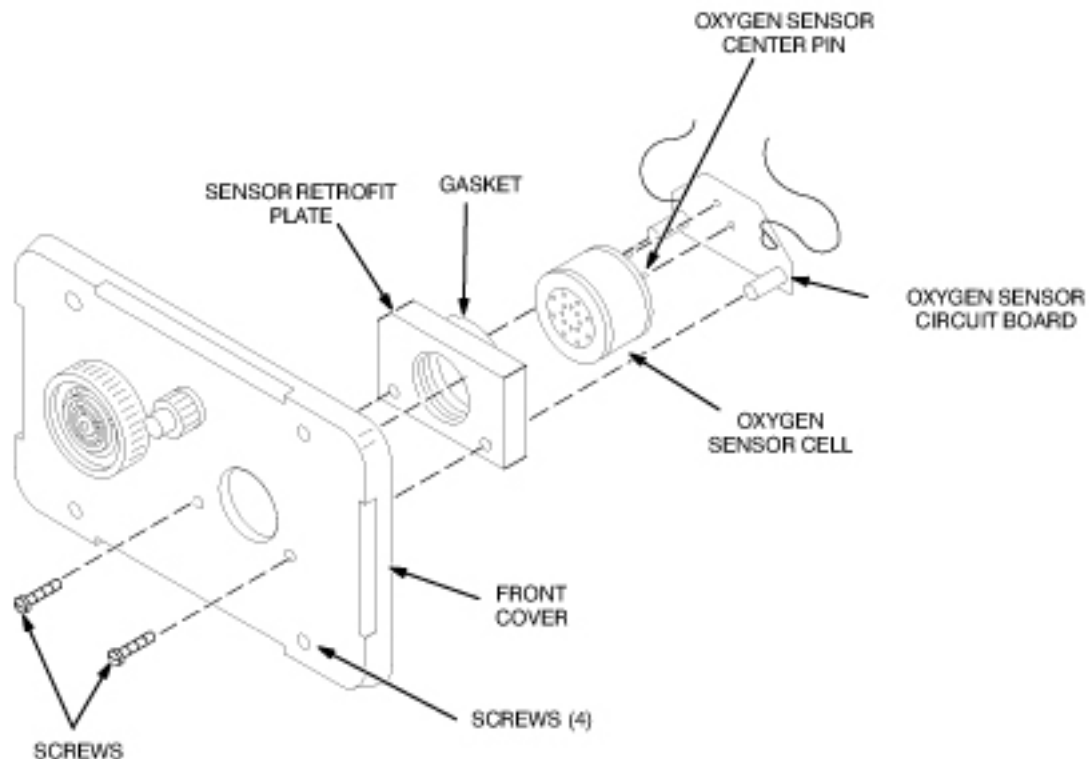
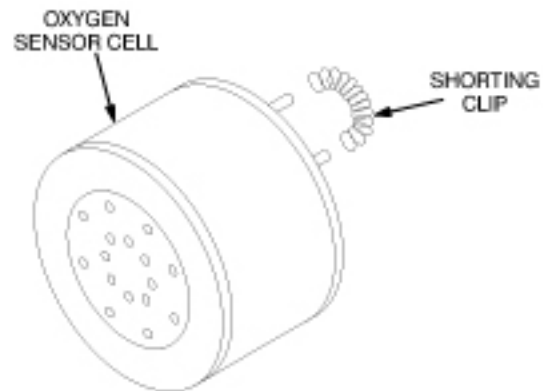


Illustration 6-7: Unpacking The Oxygen Sensor Cell



1.2 Model ISA-60M

See Vendor Manual, part number 80003-500 for the replacement processes for the ISA-60M Oxygen Monitor.

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Chapter 7 Renewal Parts

1 Renewal Parts

1.1 Model ISA-42M

NOTE: See [Section 1.2](#) for renewal parts for the ISA-60M Oxygen Monitor.

1.1.1 Oxygen Monitor Kit With Cables And Remote Sensor 46-317271G1

Illustration 7-1: Oxygen Monitor Kit With Cables and Remote Sensor

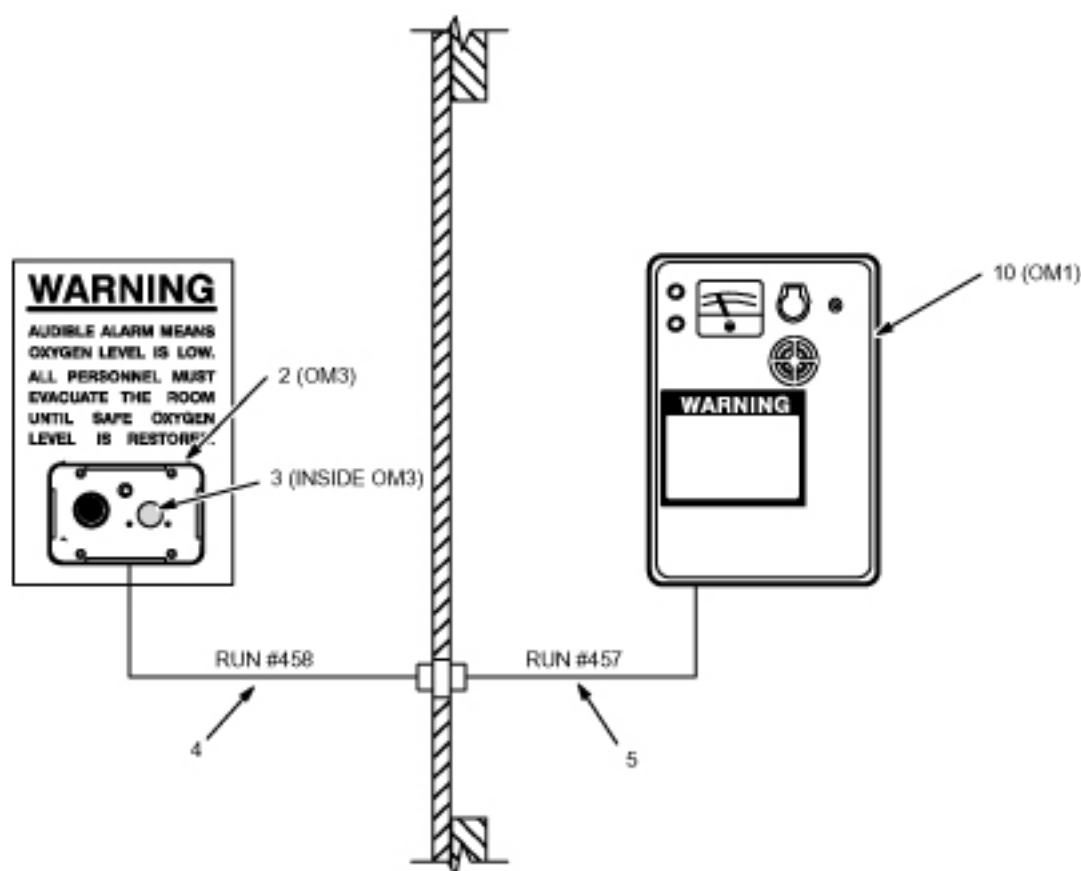


Table 7-1: Oxygen Monitor Kit with Cables and Remote Sensor

Item	Part Number	FRU	Name	Qty	Description (Remarks)
1	46-301672P2	N	O2 MON PARTS	1	REMOTE SENSOR MODULE, OM3 (SEE Section 1.1.3.)
2	46-301572P3	2	O2 MON PARTS	1	SENSOR AND MOUNTING BOARD KIT (SEE Section 1.1.3.)
3	46-243775G707	2	CABLE	1	(458) PP-1J17 TO OM3
4	46-317992G1	2	CABLE ASM	1	(457) PP-1J17 TO OM1-TB4

Item	Part Number	FRU	Name	Qty	Description (Remarks)
5	46-306515G2	2	O2 MONITOR	1	OXYGEN MONITOR, OM1 WITH RATING PLATE ATTACHED TO ENMET'S MONITOR OF THE NO OXYGEN SENSOR OR HORN SWITCH CONFIGURATION. (SEE Section 1.1.2.)

1.1.2 Oxygen Monitor (OM1) 46-306515G1

Illustration 7-2: Oxygen Monitor (OM1)

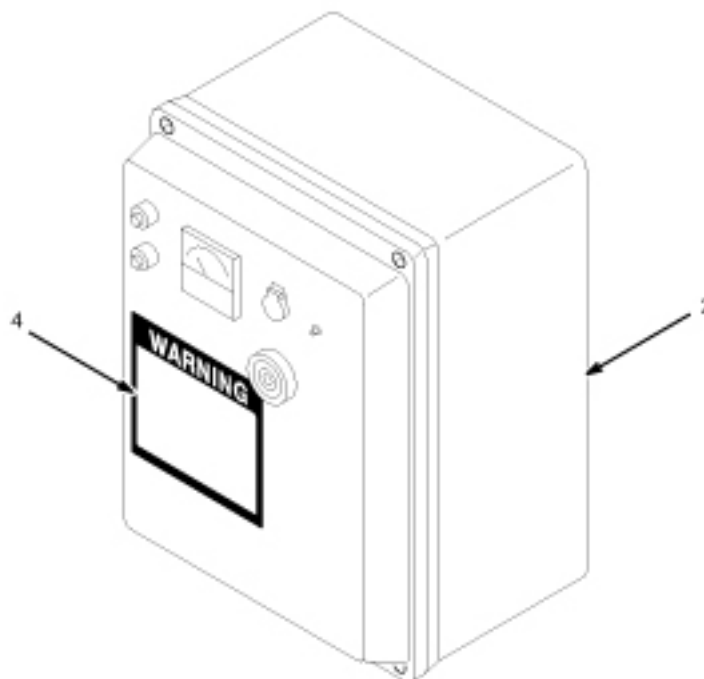
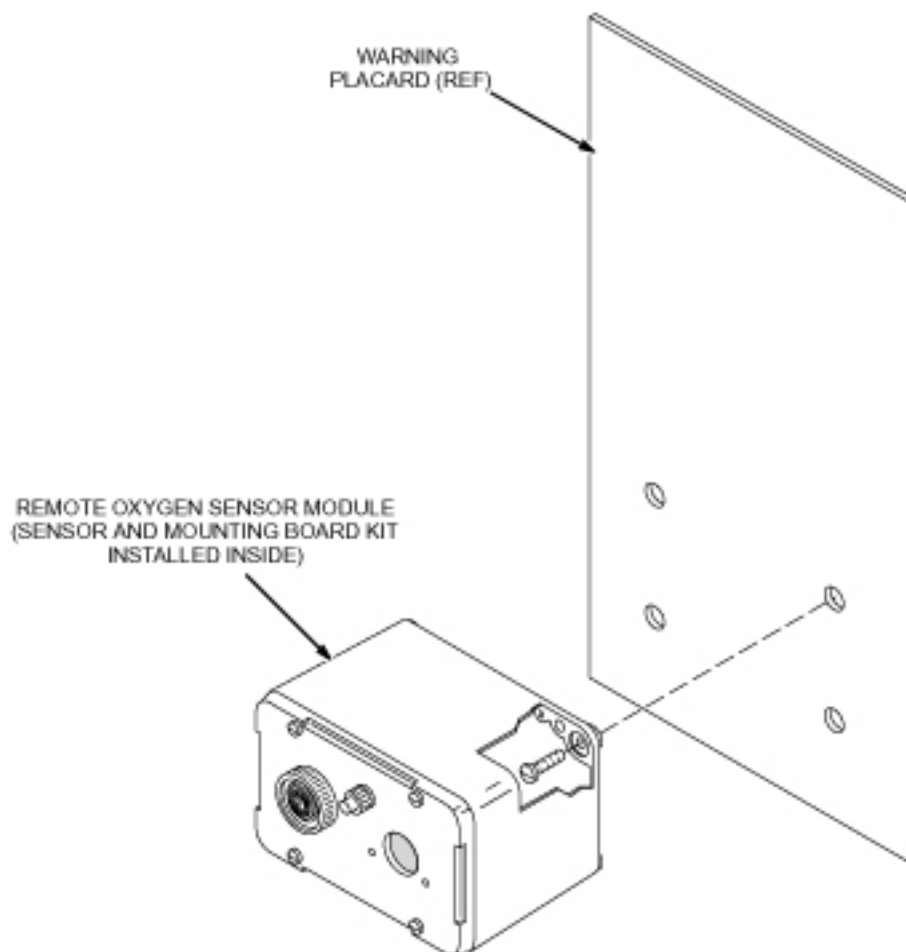


Table 7-2: Oxygen Monitor (OM1) 46-306515G1

Item	Part No.	FRU	Name	Qty	Description (Remarks)
1	46301672P1	N	O2 MON PARTS	1	ENMET OXYGEN MONITOR (SEE ENMET OXYGEN MONITOR PARTS LIST BELOW)
2	46301696P2	2	LABEL	1	WARNING LABEL, ADHESIVE BACKED

Table 7-3: ENMET Oxygen Monitor 46-301672P1

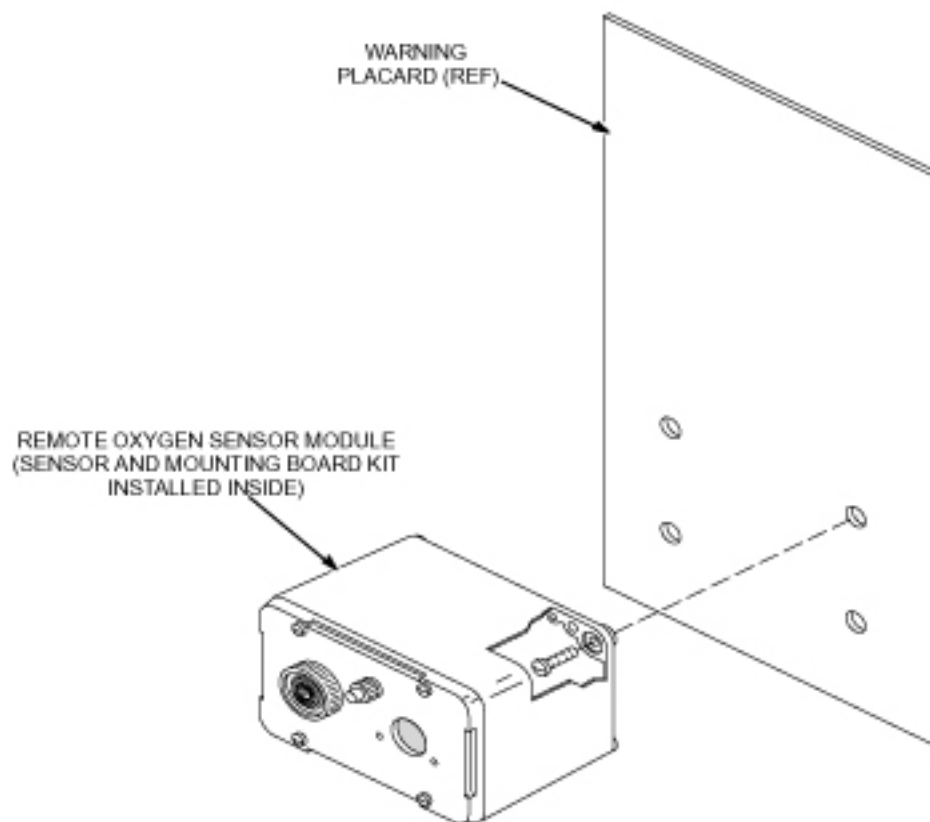
Item	Part No.	FRU	Name	Qty	Description (Remarks)
1	46-305001P1	2	ENMET PART	1	64002-050 SLOW-BLOW 0.05A FUSE
2	46-305001P3	2	ENMET PART	1	04537-000 CIRCUIT ASM
3	46-305001P5	2	ENMET PART	1	62034-001 RED LAMP
4	46-305001P6	2	ENMET PART	1	62034-002 GREEN LAMP
5	46-305001P7	2	ENMET PART	2	2034-000 LAMP BASE
6	46-305001P4	2	ENMET PART	1	62013-002 SONALERT

1.1.3 Remote Oxygen Sensor Module (OM3) 46-301672P2 (18-Month)**Illustration 7-3: Remote Oxygen Sensor Module (OM3)****Table 7-4: Remote Oxygen Sensor Module (OM3) (18-Month)**

Item	Part No.	FRU	Name	Qty	Description (Remarks)
1	46-305001P4	2	ENMET PART	1	62013-002 SONALERT
2	46-305001P5	2	ENMET PART	1	62034-001 RED LAMP
3	46-305001P7	2	ENMET PART	1	62034-000 LAMP BASE

Table 7-5: Sensor And Mounting Board Kit

Item	Part No.	FRU	Name	Qty	Description (Remarks)
1	46-305001P2	2	ENMET PART	1	67013-010 SENSOR CELL

1.1.4 Remote Oxygen Sensor Module (OM3) 46-301672P3 (5-Year)**Illustration 7-4: Remote Oxygen Sensor Module (OM3)****Table 7-6: Remote Oxygen Sensor Module (OM3) 46-301672P3 (5-Year)**

Item	Part No.	FRU	Name	Qty	Description (Remarks)
1	46-305001P4	2	ENMET PART	1	62013-002 SONALERT
2	46-305001P5	2	ENMET PART	1	62034-001 RED LAMP
3	46-305001P7	2	ENMET PART	1	62034-000 LAMP BASE

Table 7-7: Sensor And Mounting Board Kit

Item	Part No.	FRU	Name	Qty	Description (Remarks)
1	46-320936P1	2	ENMET PART	1	SENSOR RETROFIT KIT
2	5339912	2	CAC PART	1	5-Year SENSOR CELL

Table 7-8: Calibration Kit

Item	Part No.	FRU	Name	Qty	Description (Remarks)
1	5399042	1	Calibration Kit	1	Universal Calibration Kit

1.2 Model ISA-60M

The following parts are available:

Part Number	Part Number	Description
5399042	Oxygen Monitor Calibration Kit	Contains all of the parts needed to calibrate either the new or old O2 Monitor.
5390676	Oxygen Monitor FRU	Contains all of the parts included with the O2 Monitor, Box, Sensor and Cables.
5390678-3	Remote 5-year Sensor	Sensor for the ISA-60M O2 Monitor.

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